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Rebecca Kanter, Meredith P Fort
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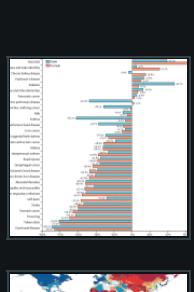
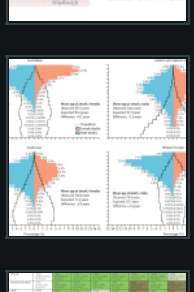
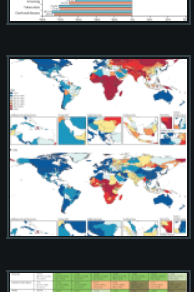
Outline

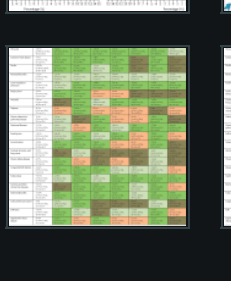
Summary
Introduction
Methods
Results
Discussion
Declaration of interests
Acknowledgments
Supplementary Material
References
Show full outline

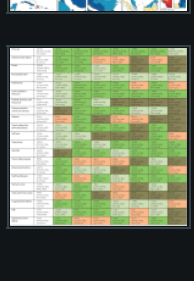
Figures (6)











Tables (2)

Summary

Background

Timely and comprehensive analyses of causes of death stratified by age, sex, and location are essential for shaping effective health policies aimed at reducing global mortality. The Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) 2023 provides cause-specific mortality estimates measured in counts, rates, and years of life lost (YLLs). GBD 2023 aimed to enhance our understanding of the relationship between age and cause of death by quantifying the probability of dying before age 70 years (70q0) and the mean age at death by cause and sex. This study enables comparisons of the impact of causes of death over time, offering a deeper understanding of how these causes affect global populations.

Methods

GBD 2023 produced estimates for 292 causes of death disaggregated by age-sex-location-year in 204 countries and territories and 660 subnational locations for each year from 1990 until 2023. We used a modelling tool developed for GBD, the Cause of Death Ensemble model (CODEm), to estimate cause-specific death rates for most causes. We computed YLLs as the product of the number of deaths for each cause-age-sex-location-year and the standard life expectancy at each age. Probability of death was calculated as the chance of dying from a given cause in a specific age period, for a specific population. Mean age at death was calculated by first assigning the midpoint age of each age group for every death, followed by computing the mean of all midpoint ages across all deaths attributed to a given cause. We used GBD death estimates to calculate the observed mean age at death and to model the expected mean age across causes, sexes, years, and locations. The expected mean age reflects the expected mean age at death for individuals within a population, based on global mortality rates and the population's age structure. Comparatively, the observed mean age represents the actual mean age at death, influenced by all factors unique to a location-specific population, including its age structure. As part of the modelling process, uncertainty intervals (UIs) were generated using the 2·5th and 97·5th percentiles from a 250-draw distribution for each metric. Findings are reported as counts and age-standardised rates. Methodological improvements for cause-of-death estimates in GBD 2023 include a correction for the misclassification of deaths due to COVID-19, updates to the method used to estimate COVID-19, and updates to the CODEm modelling framework. This analysis used 55 761 data sources, including vital registration and verbal autopsy data as well as data from surveys, censuses, surveillance systems, and cancer registries, among others. For GBD 2023, there were 312 new country-years of vital registration cause-of-death data, 3 country-years of surveillance data, 51 country-years of verbal autopsy data, and 144 country-years of other data types that were added to those used in previous GBD rounds.

Findings

The initial years of the COVID-19 pandemic caused shifts in long-standing rankings of the leading causes of global deaths: it ranked as the number one age-standardised cause of death at Level 3 of the GBD cause classification hierarchy in 2021. By 2023, COVID-19 dropped to the 20th place among the leading global causes, returning the rankings of the leading two causes to those typical across the time series (ie, ischaemic heart disease and stroke). While ischaemic heart disease and stroke persist as leading causes of death, there has been progress in reducing their age-standardised mortality rates globally. Four other leading causes have also shown large declines in global age-standardised mortality rates across the study period: diarrhoeal diseases, tuberculosis, stomach cancer, and measles. Other causes of death showed disparate patterns between sexes, notably for deaths from conflict and terrorism in some locations. A large reduction in age-standardised rates of YLLs occurred for neonatal disorders. Despite this, neonatal disorders remained the leading cause of global YLLs over the period studied, except in 2021, when COVID-19 was temporarily the leading cause. Compared to 1990, there has been a considerable reduction in total YLLs in many vaccine-preventable diseases, most notably diphtheria, pertussis, tetanus, and measles. In addition, this study quantified the mean age at death for all-cause mortality and cause-specific mortality and found noticeable variation by sex and location. The global all-cause mean age at death increased from 46·8 years (95% UI 46·6–47·0) in 1990 to 63·4 years (63·1–63·7) in 2023. For males, mean age increased from 45·4 years (45·1–45·7) to 61·2 years (60·7–61·6), and for females it increased from 48·5 years (48·1–48·8) to 65·9 years (65·5–66·3), from 1990 to 2023. The highest all-cause mean age at death in 2023 was found in the high-income super-region, where the mean age for females reached 80·9 years (80·9–81·0) and for males 74·8 years (74·8–74·9). By comparison, the lowest all-cause mean age at death occurred in sub-Saharan Africa, where it was 38·0 years (37·5–38·4) for females and 35·6 years (35·2–35·9) for males in 2023. Lastly, our study found that all-cause 70q0 decreased across each GBD super-region and region from 2000 to 2023, although with large variability between them. For females, we found that 70q0 notably increased from drug use disorders and conflict and terrorism. Leading causes that increased 70q0 for males also included drug use disorders, as well as diabetes. In sub-Saharan Africa, there was an increase in 70q0 for many non-communicable diseases (NCDs). Additionally, the mean age at death from NCDs was lower than the expected mean age at death for this super-region. By comparison, there was an increase in 70q0 for drug use disorders in the high-income super-region, which also had an observed mean age at death lower than the expected value.

Interpretation

We examined global mortality patterns over the past three decades, highlighting—with enhanced estimation methods—the impacts of major events such as the COVID-19 pandemic, in addition to broader trends such as increasing NCDs in low-income regions that reflect ongoing shifts in the global epidemiological transition. This study also delves into premature mortality patterns, exploring the interplay between age and causes of death and deepening our understanding of where targeted resources could be applied to further reduce preventable sources of mortality. We provide essential insights into global and regional health disparities, identifying locations in need of targeted interventions to address both communicable and non-communicable diseases. There is an ever-present need for strengthened health-care systems that are resilient to future pandemics and the shifting burden of disease, particularly among ageing populations in regions with high mortality rates. Robust estimates of causes of death are increasingly essential to inform health priorities and guide efforts toward achieving global health equity. The need for global collaboration to reduce preventable mortality is more important than ever, as shifting burdens of disease are affecting all nations, albeit at different paces and scales.

Funding

Gates Foundation.

Previous article in this issue Next article in this issue

Introduction

Measuring causes of death is a foundational step towards developing effective strategies to improve human health. The Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) provides comprehensive and systematic analyses of causes of death worldwide and across time. The utility of GBD cause of death estimates has been particularly valuable during the onset of COVID-19.[1], [2], [3] However, GBD estimates have uses beyond informing preparation for stochastic events, such as a novel virus or new pandemic; these estimates are used integrally as tools for understanding public health trends, shaping health policy, and monitoring progress toward global health goals.[4] As a global public good, GBD 2023 contributes freely available, updated, and comprehensive estimates of causes of death to the existing body of scientific literature. In addition to presenting the routinely updated estimates of causes of death, the current study expands our analysis to further explore the relationship between age and cause of death.

This study investigates important age patterns in mortality by estimating the probability of dying from any given cause before the age of 70 years (70q0). The probability of death measure is a fundamental indicator in public health because it can effectively capture improvements in survival within all age groups before age 70 years.[5] In recent publications, it has become common practice to classify deaths occurring before 70 years of age as premature.[6] Some studies, including the Global Health 2050 report from the *Lancet* Commission on Investing in Health, have shown that the probability of all-cause mortality before age 70 years has decreased globally and across major regions.[5] The Global Health 2050 report concluded that further reductions, by as much as 50%, are attainable by mid-century with targeted health investments, a goal referred to as 50 by 50.[5] To support progress towards 50 by 50, our study aims to address remaining questions, including which causes of death deviate from the broader improvements in premature mortality, and where disparities might exist in the likelihood of dying before age 70 years within specific populations. These are pressing concerns for policy makers and health-planning teams at both national and international levels.

Another primary objective of GBD 2023 was to calculate the mean age at the time of death across causes and locations. This metric allows for straightforward observations of national and regional disease burdens in relation to global values. Related studies find that the overall mortality rate from all causes has been decreasing over the past 75 years,[7] and the mean age at the time of death has been trending upward for many countries.[8] Estimates of mean age at death are influenced not only by a population's age distribution but also by disease characteristics, health-care access, socioeconomic status, comorbidities, and other risk factors.[8], [9] Although some of this general upward trend can be attributed to shifts in age structure and sex distribution by location, in certain areas and for specific causes, the mean age is much higher or lower than expected.[8] Quantifying the difference between the expected mean age at death (based only on population age structure and disease characteristics) and the observed mean age at death provides policy makers with additional population-level understanding beyond simply comparing age-standardised death rates between locations. These quantified differences could be linked to modifiable factors within a community, such as high blood pressure or the use of alcohol, tobacco, or drugs, which can be targeted through public health interventions.[10]

> Research in context
>
> **Evidence before this study**
>
> The Global Burden of Diseases, Injuries, and Risk Factors Study (GBD) is a worldwide research initiative that provides comprehensive and timely assessments of mortality, morbidity, and risk factors disaggregated to granular levels that are meaningful for policy development. In the last iteration, the GBD 2021 causes-of-death publication marked a major advancement in the evidence base; the study delineated cause-specific mortality to provide insights on the primary causes of death influencing life expectancy across locations. It also identified several causes with shifting mortality trends that had important implications for targeted policy initiatives—causes that were once widespread across the globe but became increasingly localised and in need of tailored reduction strategies. The GBD 2021 causes-of-death analysis was also the first of its kind to publish worldwide estimates of deaths from the initial years of the COVID-19 pandemic, quantifying its effect on life expectancy and offering comparisons to deaths from other causes. While estimates from other studies are published periodically that assess specific causes of death among a subset of populations or across a narrower timeframe, GBD remains the only research effort to offer cause-specific estimates of mortality to this degree of time and location detail and to produce these assessments in peer-reviewed and GATHER-compliant publications.
>
> **Added value of this study**
>
> This study provides new and more robust evidence of mortality patterns across the globe, updating and extending the analysis from GBD 2021, and reanalysing the entire time series to supersede all previous GBD publications. We provide estimates of cause-specific mortality for 292 causes of death within 204 countries and territories and 660 subnational locations, disaggregated by age and sex, from 1990 to 2023. These estimates include 11 474 new sources compared with GBD 2021. This update advances mortality measurements in several ways. First, we present the probability of death before age 70 years (70q0) by sex and year to enable measurements of premature mortality by individual causes. We describe causes of death that are not following global improvements in 70q0 to highlight locations where disparities are occurring in the likelihood of dying before age 70 years. Second, we calculate the mean age at death by assigning the midpoint age of each age group for every death, followed by computing the overall mean across all deaths attributed to a given cause. Our analysis of mean age of death offers insights into a country's ability to manage different disease burdens relative to global benchmarks, independent of local population structure. Additionally, our study examines the correlation between mean age at death and the Socio-demographic Index (SDI) to evaluate whether countries at the higher end of the SDI exhibit older mean ages at death for a given cause compared with countries with a lower SDI value, while controlling for SDI's effect on population structure. This approach adds a novel dimension to understanding how sociodemographic factors influence both the risk and timing of mortality. This study also builds upon our estimates from GBD 2021 to include 2 additional years of COVID-19 analysis, providing a more comprehensive picture of COVID-19 mortality worldwide. Lastly, we report estimates for several newly disaggregated causes of death, including ulcerative colitis; Crohn's disease; thyroid disease; other endocrine, metabolic, and blood and immune disease; and electrocution.
>
> **Implications of all the available evidence**
>
> Our study offers a thorough analysis of causes of death over the past 34 years, including new findings into the full duration of the COVID-19 pandemic. We highlight causes of death that have declined in certain locations, which could lend insight for policy change and implementation. We also identify causes that persist as major sources of mortality across populations, signifying priority areas for future intervention. Additionally, our study investigated important age patterns in mortality by estimating the probability of dying from any given cause before age 70 years, thereby advancing our understanding of the relationship between age and cause of death. The Global Health 2050 report set a target to reduce the probability of premature deaths by 50% by 2050. We aim to complement and support this goal by offering an in-depth analysis of 70q0 across time, sex, and geographical locations. Lastly, our mean age of death analysis is a valuable metric for comparing observed mortality levels with expected patterns to help identify locations that are keeping pace with development trends and those that might be falling behind. Evidence from this study can be used to examine epidemiological patterns and trends across time and locations, and to gauge progress in global development goals. These findings can also guide future policy initiatives aimed at furthering reductions in cause-specific mortality and, in particular, achieving better pandemic preparedness within the context of specific locations. In aggregate, cyclical updates to GBD reflect improvements in data availability and enhanced methodology that reduce bias and improve transparency, supporting the development and implementation of new evidence-based health policies worldwide.

The timeframe of this analysis allows for important new insights into COVID-19, including two additional years of estimation since GBD 2021, new data collected, and improved methodology. As we mark 5 years since the onset of the COVID-19 pandemic—declared officially by WHO in March, 2020[11]—it is important to reflect on its impact. Substantial declines in deaths from COVID-19 were not noted until 2023, after a period of extraordinary global disruption.[7] Since that time, countries with robust vital registration systems have been able to publish mortality data for the years with the highest number of COVID-19 deaths. In addition, localised studies revealed shifts in mortality patterns occurring for certain causes of death during the height of the pandemic.[12] As additional vital registration data become available, a more comprehensive understanding of the long-term effects of COVID-19 on global mortality will continue to unfold. Key questions remain regarding the total number of deaths attributed to COVID-19, the populations most affected, and which causes of death—and to what extent—were affected by the COVID-19 pandemic.

This study provides new insights related to trends in 70q0 and the mean age at death, and identifies and delineates causes that most heavily affect mortality across populations. An updated understanding of how the COVID-19 pandemic interrupted or altered previous trajectories in mortality by cause, age, sex, or location is another important contribution of GBD 2023. At the same time, tracking changes in 70q0 and the mean age at death across causes, populations, and over time—alongside metrics such as the number of deaths, age-standardised mortality rates, and years of life lost (YLLs), offers more actionable insights to improve health at the population level. These patterns can be an essential guide for policy makers when shaping health priorities. This manuscript was produced as part of the GBD Collaborator Network and in accordance with the GBD Protocol.[13]

Methods

Overview

GBD 2023 produced estimates for each epidemiological quantity of interest for 292 causes of death by age-sex-location-year for 25 age groups from birth to 95 years and older; for males, females, and all sexes combined; in 204 countries and territories grouped into 21 regions and seven super-regions; and for every year from 1990 to 2023. GBD 2023 also includes subnational analyses for 20 countries and territories. This study drew on the expertise of a network of 14 410 international collaborators from more than 160 countries and territories who provide, review, and analyse the available data to generate these metrics.

GBD 2023 produced updated estimates of health loss around the world using the best available data. For each GBD round, newly available data and updated methods are used to update the full time series of estimates from 1990 to the latest year of analysis. Consequently, GBD 2023 estimates supersede all previous estimates. The methods used to generate estimates for GBD 2023 closely followed those for GBD 2021.[14] These methods have been extensively peer reviewed over previous rounds of GBD[14], [15], [16], [17], [18], [19] and concurrently as part of the peer review process for GBD 2023. Here, we provide an overview of the methods with an emphasis on the main methodological changes since GBD 2021; a comprehensive description of the analytical methods for GBD 2023 is provided in appendix 1.

The GBD 2023 cause-of-death estimates described here include cause-specific mortality, observed and expected mean ages at death, cause-specific probabilities of death before age 70 years, and the premature death metric YLLs. YLLs were calculated as the number of deaths for each cause-age-sex-location-year multiplied by the standard life expectancy at each age (appendix 1 section 6.3). Standard life expectancy is calculated from the lowest age-specific mortality rate between countries.[7] In brief, cause-specific death rates for 214 causes were estimated using the Cause of Death Ensemble model (CODEm), while alternative strategies were used to model causes with very limited data, changes in reporting over the study period, or very specific epidemiology. The modelling strategy used for all cause of death estimates can be found in appendix 1 (table S8). CODEm is a modelling tool developed specifically for GBD that evaluates the out-of-sample predictive validity of different statistical models and covariate permutations and then combines the results from those evaluations to produce cause-specific fatal burden estimates.

Methodological improvements for cause-of-death estimates in the current round of estimation focused on several key areas. First, a method for the identification and correction of causes displaying excess mortality spikes due to misclassified COVID-19 deaths was applied to all vital registration data between the years of 2020 and 2023. Second, we added 312 new country-years of vital registration data on cause of death, 3 country-years of surveillance data, 51 country-years of verbal autopsy data, and 144 country-years of other data types. Third, all CODEms were fitted to mortality rates rather than cause fractions. Fourth, we updated the modelling framework for COVID-19 to incorporate pandemic-era vital registration data and preliminary vital registration reporting.

The GBD disease and injury hierarchy

GBD classifies diseases and injuries into a hierarchy with four Levels that include both fatal and non-fatal causes. Level 1 causes include three broad aggregate categories (communicable, maternal, neonatal, and nutritional [CMNN] diseases; non-communicable diseases [NCDs]; and injuries); Level 2 disaggregates those categories into 22 clusters of causes, which are further disaggregated into Level 3 and Level 4 causes. At the most detailed Level, 292 fatal causes are estimated. For a full list of causes of death by Level, see appendix 1 (table S1). For GBD 2023, five causes of death were estimated for the first time: ulcerative colitis; Crohn's disease; thyroid disease; other endocrine, metabolic, blood, and immune disease; and electrocution.

Data sources, processing, and assessing for completeness

The GBD 2023 cause-of-death database included data sources identified in previous rounds of estimation in addition to 11 474 new sources, for a total of 55 761 data sources—these sources are detailed in appendix 1 (table S3) and can be accessed through the Global Health Data Exchange (GHDx) website. Multiple data types were included to capture the widest array of information, including vital registration data for all 292 causes, as well as verbal autopsy, survey, census, surveillance, cancer registry, and police record data; open-source databases; and minimally invasive tissue sampling. To standardise these data so that they could be compared by cause, age, sex, location, and time, a set of data processing corrections were applied. First, deaths with insufficient or missing age and sex detail underwent a process of distribution via age and sex splitting (appendix 1 section 3.5). In addition, garbage codes, which are non-specific, implausible, or intermediate rather than underlying cause-of-death codes from the ICD, were redistributed to appropriate targets to assign the underlying cause of death.[20] Data sources with more than 50% of all deaths assigned to major garbage codes (class 1 or class 2 garbage codes) in any location-year were excluded to mitigate the potential for bias from these sources (appendix 1 section 3.11).

Assessing data completeness illustrates the coverage from a data source on overall mortality for the country. Vital registration and verbal autopsy data completeness—a source-specific estimate of the percentage of total cause-specific deaths that are reported in a given location and year—was assessed by location-year, and sources with less than 50% completeness were excluded. We excluded 283 country-years of data due to insufficient completeness or excessive garbage coding. The estimated all-cause mortality for each age-sex-location-year was then multiplied by the cause fraction for the corresponding age-sex-location-year to adjust all included sources to 100% completeness. GBD assesses the quality of all vital registration and verbal autopsy data using a star ranking system of one to five stars, based on the percentage of completeness and percentage of garbage coding. Vital registration and verbal autopsy data availability, completeness, and five-star quality rating for each location-year are available in appendix 1 (figures S4 and S5). Full details on all data processing corrections can be found in appendix 1 (section 3.16).

Presentation of cause-specific mortality estimates

Cause-specific mortality estimates for GBD 2023 are given in death counts and age-standardised rates per 100 000 population, calculated using the GBD standard population structure.[7] For changes over time, we present percentage changes over the period 1990–2023, and annualised rates of change as the difference in the natural log of the values at the start and end of the time interval divided by the number of years in the interval. 95% uncertainty intervals (UIs) for all metrics are computed using the 2·5th and 97·5th percentiles from a 250-draw distribution for each metric (appendix 1 section 4.1.3). To reduce computing power and time, the number of computations per process was scaled back from 500 in GBD 2021 to 250 in GBD 2023, as simulation testing revealed that final estimates and their uncertainty were not affected by this reduction. See appendix 1 (section 4.1.3) for further details on this update.

Measuring probability of premature death

In accordance with the GBD framework, a death that occurs at any age before the standard (expected) life expectancy is classified as premature. To inform discussions and debates in the literature on premature deaths before age 70 years, in alignment with studies from WHO,[6], [21] the US National Institutes of Health,[22] and the US Centers for Disease Control and Prevention,[23] we computed the probability of death from birth to age 70 years (70q0).

Calculation of the probability of premature death by cause

The probability-of-death metric represents the chance of dying from a given cause in a specific age period, for a specific population. Methods for calculating all-cause probability of premature death have been described elsewhere.[7] For example, for males aged 0–70 years in Canada in 1990, a probability of death of 0·1 from ischaemic heart disease indicates a 10% chance of dying from this cause before age 70 years. Cause-specific probability of death can be calculated as follows:

$$q^n_{x,c} = deaths^n_{x,c} \times q^n_x$$

where

$$q^n_{x,c}$$

represents the probability of death for ages *x* to *x* + *n* in cause *c*,

$$\mathrm{deaths}^n_{x,c}$$

represents deaths in cause fraction space for age group *x* to *x* + *n* in cause *c*; and

$$q^n_x$$

represents the all cause probability of death for ages *x* to *x* + *n*.

Socio-demographic Index

The Socio-demographic Index (SDI) is a composite measure of two demographic indicators (total fertility rate in people younger than 25 years and mean educational attainment for those aged 15 years and older) and an economic indicator (lag-distributed income per capita).[7] Values are given as a range between 0·0 and 1·0. Additional details describing the calculation of SDI for GBD 2023 are provided in appendix 1 (section 5) of our companion publication.[7]

Calculation of the observed mean age at death and the expected mean age at death based on the global age-specific rates

The calculation of mean age at death uses cause-specific GBD modelled death estimates. GBD produces cause-of-death estimates for every location-year-age-sex group, even when no reported cause-of-death data are available. GBD uses standard 5-year age groups (eg, 15–19, 20–24, and 25–29 years) from age 5 years to 94 years. The remaining non-standard age groups consist of 0–6 days, 7–27 days, 1–5 months, 6–11 months, 12–23 months, 2–4 years, and 95 years and older. For this calculation, each GBD age group is assigned a distinct age at death by taking the mean age of each age group. For example, the age group 15–19 years, which represents people from the day they turn 15 years of age to the day before they turn 20 years of age, was assigned to have a distinct age at death of 17·5 years. The only age group without a discernible mean is the 95 years and older group. For this age group, the distinct age at death was calculated by adding the life expectancy of the 95 years and older age group to 95 years by sex-year-location.

The observed mean age at death uses GBD estimates directly, and each death can be assigned a distinct age at death. Distinct ages are then summed together for a given demographic consisting of a given location-year-sex-cause. This value is then divided by the total number of deaths for the same demographic to quantify the mean age at death.

Expected deaths were calculated using cause-specific GBD global mortality rates by age and sex and applying them to each country's population. By multiplying the mortality rate on the population to calculate deaths, expected death estimates control for population structure. These expected death estimates are comparable to normal GBD estimates, consisting of the same age-sex groups. The same process used to calculate the observed mean age at death is then applied to the expected deaths to calculate the expected mean age at death.

The relationship between mean age at death and SDI was explored by running linear regressions of SDI against the observed mean age at death as well as running linear regressions of SDI against the difference of observed and expected mean ages at death. Appendix 2 (table S14) reports the resulting r^2, slope, and p values of each regression. An individual regression was run for each cause–sex combination, in which each observation represents a country in 1990, 2010, 2019, 2021, or 2023. By observing the relationship between SDI and the difference between observed and expected values, we are able to measure the correlation of SDI with the mean age at death while accounting for differences in population structure that might also be correlated with SDI.

Correction for the misclassification of COVID-19 deaths

GBD 2023 received 83 country-years of vital registration data from 2020, 67 from 2021, and 38 from 2022. During these years, there is evidence that deaths due to COVID-19 were misclassified as other causes of death.[24], [25] Relative to smooth prepandemic mortality trends, these misclassified COVID-19 deaths contributed to mortality spikes in the other causes of death. To systematically identify these spikes in other causes, we developed a Support Vector Machine, a machine-learning algorithm for identifying deviations from the established time trends in the years 2020–22. After we identified causes with mortality spikes during the COVID-19 pandemic years, we ascertained, for each cause of interest, whether the spike was a result of COVID-19 misclassification or a true increase in mortality. To do this, we evaluated the correlation between the rate of excess mortality in the cause of interest and the observed mortality rate of COVID-19. Mortality spikes identified to contain COVID-19 misclassification were then considered eligible for correction.

When a mortality spike had been identified as being eligible for correction, we calculated the portion of excess mortality attributable to misclassified COVID-19. We first created an estimate of expected deaths absent of any pandemic effects using the mean of two counterfactual estimates: one calculated by a linear regression of the 5 years before the start of the pandemic (2015–19), and another calculated using a global relative rate of non-COVID-19 deaths, adapted from a previously published method used in the correction of misclassified HIV.[26] Total excess mortality could then be estimated by subtracting the expected death count total from the observed death count total. Finally, total excess mortality was then scaled according to the level of correlation between COVID-19 rate and excess mortality rate to calculate the amount of excess attributable to COVID-19. The total excess attributable to COVID-19 was then subtracted from the cause of interest and reassigned to COVID-19. The full details regarding the identification and correction of misclassified COVID-19 can be found in appendix 1 (section 3.8), appendix 2 (table S2), and appendix 3. Detailed results of this process can be found in appendix 2 (table S3).

Estimation of COVID-19 as a cause of death

For modelling COVID-19, we supplemented the corrected vital registration data described above with two other sources of data: 9 country-years of provisional vital registration data and 342 country-years of surveillance data that were reported during the pandemic (to 2022). We developed an analysis method using OneMod, a modelling tool that combines robust feature selection, correlated time-series splines, and covariate effect sizes across age groups, in addition to kernel regression for residual smoothing. It included the following candidate covariates: total COVID-19 infections and variant prevalence; COVID-19 vaccinations;[27] COVID-19 infection detection rate;[27] Healthcare Access and Quality Index;[28] and prevalence of risk factors and comorbidities including obesity, smoking, cancer, cardiovascular disease, chronic kidney disease, chronic obstructive pulmonary disease, and diabetes.29, 30 In the first stage of this model pipeline, we used only the corrected vital registration data to estimate age patterns and sex ratios, which were then used to split the provisional vital registration by age and sex, and split the surveillance data that did not contain detailed age and sex information into the 25 granular GBD age groups. We then ran the models using the entire dataset, setting the infection detection rate to 100% for the corrected vital registration data. After fitting these models, we made predictions assuming that the infection detection rate was 100% in all locations. Details on the estimation of COVID-19 deaths can be found in appendix 1 (section 5).

GBD research and reporting practices

This study used de-identified data and was approved by the University of Washington Institutional Review Board (study number 9060). GBD 2023 complies with the Guidelines for Accurate and Transparent Health Estimates Reporting (GATHER) statement (appendix 1 section 2.4).[31] A completed GATHER checklist is provided in appendix 1 (table S13). Software packages used in the cause of death analysis for GBD 2023 were Python version 3.10.4, Stata version 13.1, and R version 4.4.0. Statistical code used for GBD estimation is publicly available online at the GHDx website ↗.

Role of the funding source

The funders of this study had no role in the study design, data collection, data analysis, data interpretation, or the writing of the report. The corresponding author had full access to the data in the study and final responsibility for the decision to submit for publication.

Results

Detailed results for each cause of death in this analysis are available in downloadable form through the GBD Results tool ↗ and via visual exploration through the GBD Compare tool ↗.

Global all-cause mortality

Relative to the rate in 1990, the percentage change in annual age-standardised mortality rate from 1991–2019 globally for all causes of death fluctuated between a decrease of 2·9% (95% UI −3·5 to −2·3) and a slight increase of 0·1% (−0·7 to 1·0; appendix 2 figure S1). A notable increase occurred between 2019 and 2020 (6·5% [5·9 to 7·1]) and 2020 and 2021 (7·0% [6·5 to 7·6]), followed by a large decrease between 2021 and 2022 (−9·5% [−10·7 to −8·2]). The total number of global deaths for all sexes and all age groups increased from 47·9 million (47·6–48·3) in 1990 to 55·2 million (54·8–55·5) in 2019. An increase occurred during the initial years of the COVID-19 pandemic, with global deaths reaching 65·9 million (65·6–66·2) in 2021. As the pandemic subsided, the annual global death toll decreased to 60·0 million (59·0–61·2) in 2023. From 2000 to 2023, this represents an overall decline of 30·5% (29·0–31·7), in which the global age-standardised mortality rate for all sexes and age groups dropped from 1009·0 (1002·5–1015·2) deaths per 100 000 to 701·5 (690·2–714·9) deaths per 100 000 (table 1, appendix 2 table S7).

Table 1. Global death and YLL numbers, age-standardised rates per 100 000, and percentage change between 2000 and 2023 for all sexes combined for all GBD causes and Levels 1–4 of the cause hierarchy

	All-age deaths, thousands	
	2000	**2023**
All causes	**50 746·1 (50 430·8 to 51 060·8)**	**60 043·1 (59 045·4**
Communicable, maternal, neonatal, and nutritional diseases	**15 054·6 (14 495·6 to 15 946·2)**	**10 326·0 (9761·6 to**
HIV/AIDS and sexually transmitted infections	**1578·2 (1462·9 to 1719·2)**	**917·2 (799·7 to 107**
HIV/AIDS	1489·3 (1387·8 to 1620·1)	833·4 (727·1 to 959
HIV/AIDS and drug-susceptible tuberculosis co-infection	477·2 (340·6 to 569·7)	190·5 (125·2 to 252
HIV/AIDS and multidrug-resistant tuberculosis without extensive drug resistance co-infection	29·9 (7·8 to 81·5)	18·8 (6·5 to 38·9)
HIV/AIDS and extensively drug-resistant tuberculosis co-infection	0·4 (0·1 to 0·9)	0·8 (0·3 to 1·6)
HIV/AIDS resulting in other diseases	981·9 (863·6 to 1174·0)	623·3 (519·2 to 770
Sexually transmitted infections excluding HIV	88·9 (39·6 to 164·0)	83·7 (35·7 to 154·3)
Syphilis	82·9 (33·2 to 159·9)	77·0 (29·5 to 147·5)
Chlamydial infection	1·3 (0·7 to 2·6)	1·5 (0·9 to 2·6)
Gonococcal infection	0·6 (0·3 to 1·0)	0·6 (0·4 to 1·0)
Other sexually transmitted infections	4·0 (2·3 to 7·7)	4·6 (2·8 to 8·0)
Respiratory infections and tuberculosis	**4446·2 (4000·3 to 4975·1)**	**4337·1 (3947·6 to 4**
Tuberculosis	1760·3 (1397·6 to 2165·4)	1010·5 (806·6 to 12
Drug-susceptible tuberculosis	1629·6 (1241·5 to 2010·7)	908·3 (670·6 to 114
Multidrug-resistant tuberculosis without extensive drug resistance	127·1 (35·2 to 305·7)	95·5 (28·9 to 224·1)
Extensively drug-resistant tuberculosis	3·6 (1·1 to 8·8)	6·8 (2·6 to 14·5)
Lower respiratory infections	2646·6 (2369·3 to 2950·0)	2501·3 (2241·0 to 2
Upper respiratory infections	38·3 (7·1 to 93·9)	27·1 (6·1 to 77·3)
Otitis media	1·0 (0·3 to 3·7)	0·7 (0·3 to 1·7)
COVID-19	0·0 (0·0 to 0·0)	797·6 (722·9 to 857
Enteric infections	**2658·5 (2142·0 to 3452·1)**	**1268·6 (962·6 to 16**
Diarrhoeal diseases	2336·2 (1838·9 to 3112·9)	1107·1 (810·5 to 15
Typhoid and paratyphoid	231·3 (122·0 to 378·2)	82·8 (43·4 to 135·4)
Typhoid fever	196·9 (101·7 to 326·2)	72·0 (38·1 to 118·6)
Paratyphoid fever	34·4 (16·0 to 64·7)	10·9 (5·2 to 20·9)
Invasive non-typhoidal salmonella	89·5 (72·5 to 112·2)	76·7 (60·1 to 100·1)
Other intestinal infectious diseases	1·5 (1·2 to 1·9)	1·9 (1·4 to 2·4)
Neglected tropical diseases and malaria	**1050·3 (652·3 to 1601·1)**	**804·9 (394·2 to 13**
Malaria	881·4 (478·4 to 1450·7)	670·0 (261·2 to 125
Chagas disease	10·6 (9·8 to 11·6)	8·4 (7·5 to 9·4)
Leishmaniasis	10·1 (2·5 to 24·2)	4·6 (1·9 to 8·7)
Visceral leishmaniasis	10·1 (2·5 to 24·2)	4·6 (1·9 to 8·7)
African trypanosomiasis	26·5 (13·4 to 45·8)	1·4 (0·7 to 2·5)
Schistosomiasis	21·6 (19·8 to 23·9)	13·5 (12·3 to 14·8)
Cysticercosis	2·4 (1·8 to 3·2)	1·5 (1·1 to 2·1)
Cystic echinococcosis	3·0 (2·3 to 3·8)	1·4 (1·0 to 1·8)
Dengue	24·0 (9·0 to 60·2)	52·7 (25·2 to 108·9)
Yellow fever	12·6 (4·5 to 26·7)	4·4 (1·5 to 9·3)
Rabies	26·5 (13·9 to 44·0)	15·8 (6·7 to 27·4)
Intestinal nematode infections	14·3 (10·9 to 18·9)	5·0 (3·8 to 6·1)
Ascariasis	14·3 (10·9 to 18·9)	5·0 (3·8 to 6·1)
Ebola virus disease	0·3 (0·3 to 0·4)	0·0 (0·0 to 0·0)
Zika virus disease	0·0 (0·0 to 0·0)	0·0 (0·0 to 0·0)
Other neglected tropical diseases	16·9 (9·9 to 29·1)	26·3 (13·0 to 48·3)
Other infectious diseases	**1888·8 (1440·7 to 2382·5)**	**841·7 (682·3 to 100**
Meningitis	428·9 (341·9 to 547·5)	258·8 (202·2 to 334
Encephalitis	82·0 (52·0 to 119·1)	76·5 (54·7 to 113·4)
Diphtheria	23·3 (18·0 to 29·6)	4·0 (3·0 to 5·3)
Pertussis	205·9 (111·4 to 328·6)	115·2 (66·6 to 189·2
Tetanus	126·4 (91·1 to 179·2)	19·8 (11·7 to 31·0)
Measles	762·8 (347·4 to 1338·5)	143·6 (58·6 to 255·2
Varicella and herpes zoster	15·3 (14·0 to 16·5)	13·7 (12·2 to 14·8)
Acute hepatitis	171·7 (124·5 to 227·8)	93·1 (65·4 to 123·4)
Acute hepatitis A	103·8 (75·1 to 147·2)	35·6 (21·8 to 52·0)
Acute hepatitis B	54·1 (31·1 to 87·3)	45·9 (28·0 to 66·9)
Acute hepatitis C	10·4 (5·0 to 18·1)	7·2 (3·8 to 11·9)
Acute hepatitis E	3·5 (1·6 to 6·5)	4·4 (2·1 to 8·1)
Other unspecified infectious diseases	72·5 (41·4 to 115·1)	117·0 (65·3 to 191·9
Maternal and neonatal disorders	**3007·2 (2877·3 to 3149·7)**	**1867·9 (1739·6 to 1**
Maternal disorders	396·7 (355·3 to 438·6)	239·9 (207·8 to 280
Maternal haemorrhage	133·9 (102·0 to 165·4)	52·0 (35·8 to 70·0)
Maternal sepsis and other pregnancy-related infections	43·6 (30·6 to 61·2)	26·7 (19·0 to 37·1)
Maternal hypertensive disorders	68·9 (53·9 to 85·1)	48·2 (37·3 to 59·9)
Maternal obstructed labour and uterine rupture	22·1 (13·4 to 33·4)	12·2 (7·5 to 18·4)
Maternal abortive outcome	43·4 (29·1 to 64·0)	19·6 (12·2 to 30·8)
Ectopic pregnancy	9·9 (6·6 to 14·1)	12·7 (8·5 to 17·8)
Indirect maternal deaths	29·7 (19·6 to 42·7)	23·0 (16·3 to 32·7)
Late maternal deaths	7·2 (6·1 to 8·9)	8·0 (6·8 to 9·9)
Maternal deaths aggravated by HIV/AIDS	2·7 (1·7 to 3·6)	1·7 (1·0 to 2·3)
Other direct maternal disorders	35·3 (23·1 to 51·0)	35·7 (25·2 to 51·8)
Neonatal disorders	2610·4 (2495·1 to 2736·7)	1628·0 (1506·0 to 1
Neonatal preterm birth	1064·4 (922·5 to 1244·7)	623·3 (509·7 to 735
Neonatal encephalopathy due to birth asphyxia and trauma	825·2 (676·6 to 995·4)	562·1 (471·4 to 678
Neonatal sepsis and other neonatal infections	324·7 (213·8 to 466·6)	223·2 (155·6 to 315
Haemolytic disease and other neonatal jaundice	99·2 (54·4 to 180·0)	30·3 (16·7 to 47·8)
Other neonatal disorders	296·9 (207·3 to 420·9)	189·0 (124·2 to 267
Nutritional deficiencies	**425·4 (355·6 to 499·8)**	**288·6 (231·1 to 350**
Protein-energy malnutrition	378·2 (316·6 to 448·9)	245·8 (195·9 to 300
Other nutritional deficiencies	47·2 (36·0 to 61·1)	42·8 (30·1 to 58·5)
Non-communicable diseases	**31 130·5 (30 469·8 to 31 722·1)**	**44 842·6 (43 824·4**
Neoplasms	**7072·9 (6679·9 to 7356·8)**	**10 567·9 (9726·9 to**
Lip and oral cavity cancer	120·6 (108·1 to 131·0)	225·7 (198·6 to 262
Nasopharynx cancer	64·2 (56·3 to 71·5)	75·4 (63·3 to 89·1)
Other pharynx cancer	54·2 (46·7 to 65·7)	113·9 (92·5 to 140·6
Oesophageal cancer	443·5 (380·3 to 481·6)	577·8 (505·7 to 643
Stomach cancer	943·1 (808·3 to 1042·2)	935·9 (797·9 to 108
Colon and rectum cancer	704·3 (655·8 to 742·9)	1107·1 (997·7 to 12
Liver cancer	325·8 (297·8 to 357·9)	507·7 (442·4 to 570
Liver cancer due to hepatitis B	141·4 (123·5 to 162·0)	188·3 (161·3 to 220
Liver cancer due to hepatitis C	93·5 (82·0 to 107·7)	155·5 (131·1 to 186
Liver cancer due to alcohol use	51·1 (42·4 to 63·0)	95·1 (75·9 to 117·6)
Liver cancer due to NASH	20·5 (16·3 to 25·4)	41·7 (31·6 to 52·6)
Hepatoblastoma	3·9 (2·9 to 5·2)	3·5 (2·3 to 5·2)
Liver cancer due to other causes	15·5 (13·0 to 18·5)	23·6 (18·4 to 29·7)
Gallbladder and biliary tract cancer	118·6 (105·8 to 132·7)	184·1 (159·8 to 220
Pancreatic cancer	280·5 (262·5 to 294·3)	552·7 (501·8 to 588
Larynx cancer	87·9 (77·7 to 98·8)	130·8 (112·1 to 156
Tracheal, bronchus, and lung cancer	1322·0 (1243·4 to 1408·8)	2037·1 (1857·7 to 2
Malignant skin melanoma	41·9 (38·7 to 45·5)	66·6 (59·9 to 75·2)
Non-melanoma skin cancer	30·2 (27·2 to 33·7)	63·9 (54·4 to 71·6)
Non-melanoma skin cancer (squamous-cell carcinoma)	30·2 (27·2 to 33·7)	63·9 (54·4 to 71·6)
Soft tissue and other extraosseous sarcomas	37·9 (31·1 to 48·9)	60·9 (49·0 to 75·8)
Malignant neoplasm of bone and articular cartilage	49·0 (40·5 to 61·5)	76·7 (60·8 to 96·4)
Breast cancer	446·4 (408·7 to 481·8)	780·2 (685·7 to 871
Cervical cancer	235·7 (190·6 to 300·7)	369·5 (291·7 to 474
Uterine cancer	67·2 (57·6 to 74·9)	108·6 (93·2 to 126·2
Ovarian cancer	129·9 (117·1 to 142·7)	221·0 (191·6 to 255
Prostate cancer	274·4 (243·7 to 301·9)	473·0 (415·9 to 530
Testicular cancer	8·8 (7·2 to 10·9)	11·9 (9·6 to 14·7)
Kidney cancer	101·1 (93·2 to 108·7)	165·4 (146·7 to 180
Bladder cancer	146·2 (135·1 to 156·8)	233·7 (208·5 to 257
Brain and central nervous system cancer	175·2 (150·9 to 200·1)	264·2 (230·5 to 313
Eye cancer	9·5 (6·5 to 14·1)	10·1 (7·3 to 14·2)
Retinoblastoma	4·0 (1·8 to 8·4)	3·2 (1·3 to 7·0)
Other eye cancers	5·5 (4·3 to 7·1)	6·9 (5·4 to 8·8)
Neuroblastoma and other peripheral nervous cell tumours	4·2 (3·5 to 5·0)	6·0 (5·0 to 7·6)
Thyroid cancer	28·8 (25·7 to 33·5)	52·2 (44·7 to 61·5)
Mesothelioma	17·2 (15·6 to 19·2)	28·0 (24·8 to 30·9)
Hodgkin lymphoma	28·8 (22·4 to 35·4)	27·2 (21·0 to 34·9)
Non-Hodgkin lymphoma	188·5 (173·9 to 209·5)	283·1 (247·5 to 320
Burkitt lymphoma	5·0 (3·6 to 6·8)	6·7 (5·1 to 9·6)
Other non-Hodgkin lymphoma	183·5 (169·0 to 203·9)	276·4 (241·6 to 312
Multiple myeloma	68·2 (62·3 to 73·8)	125·1 (112·8 to 137
Leukaemia	291·9 (260·2 to 320·5)	342·0 (307·2 to 381
Acute lymphoid leukaemia	87·3 (64·5 to 111·6)	77·5 (54·2 to 98·7)
Chronic lymphoid leukaemia	37·5 (33·5 to 41·6)	44·8 (39·6 to 51·6)
Acute myeloid leukaemia	91·9 (75·5 to 108·0)	131·6 (112·6 to 152
Chronic myeloid leukaemia	34·3 (28·1 to 40·9)	26·3 (21·2 to 32·7)
Other leukaemia	41·1 (33·0 to 53·7)	61·7 (49·3 to 77·0)
Other malignant neoplasms	157·2 (142·0 to 174·1)	226·0 (196·9 to 251
Other neoplasms	69·9 (61·2 to 79·1)	124·5 (106·4 to 145
Myelodysplastic, myeloproliferative, and other haemopoietic neoplasms	26·1 (23·0 to 29·0)	62·0 (51·7 to 71·9)
Other benign and in-situ neoplasms	43·8 (35·8 to 52·8)	62·5 (49·1 to 79·0)
Cardiovascular diseases	**14 562·6 (13 675·2 to 15 351·9)**	**19 159·2 (17 364·3**
Rheumatic heart disease	437·4 (347·4 to 541·2)	388·9 (261·0 to 554
Ischaemic heart disease	6286·6 (5802·6 to 6713·4)	8905·9 (8043·9 to 9
Stroke	6059·2 (5579·8 to 6549·6)	6793·2 (6064·9 to 7
Ischaemic stroke	2853·9 (2602·5 to 3179·8)	3279·0 (2869·9 to 3
Intracerebral haemorrhage	2857·0 (2539·1 to 3198·6)	3156·7 (2752·8 to 3
Subarachnoid haemorrhage	348·4 (245·3 to 428·2)	357·5 (303·7 to 430
Hypertensive heart disease	797·5 (643·5 to 953·1)	1485·0 (1179·4 to 1
Non-rheumatic valvular heart disease	97·5 (85·8 to 107·2)	191·3 (157·0 to 214
Non-rheumatic calcific aortic valve disease	71·3 (61·2 to 78·4)	149·4 (120·1 to 166
Non-rheumatic degenerative mitral valve disease	25·3 (21·7 to 29·2)	39·7 (32·4 to 50·7)
Other non-rheumatic valve diseases	0·9 (0·6 to 1·6)	2·1 (1·3 to 3·3)
Cardiomyopathy and myocarditis	335·7 (295·6 to 388·6)	399·9 (338·4 to 465
Myocarditis	21·6 (14·5 to 33·6)	16·9 (11·3 to 24·1)
Alcoholic cardiomyopathy	76·5 (68·6 to 86·0)	62·3 (56·0 to 71·6)
Other cardiomyopathy	237·6 (203·3 to 287·4)	320·7 (260·3 to 380
Pulmonary arterial hypertension	20·0 (15·0 to 27·0)	22·8 (17·5 to 29·8)
Atrial fibrillation and flutter	160·4 (144·5 to 174·3)	377·7 (319·0 to 424
Aortic aneurysm	109·6 (101·9 to 119·8)	167·4 (147·1 to 187
Lower extremity peripheral arterial disease	52·9 (47·8 to 57·6)	74·9 (66·1 to 83·1)
Endocarditis	52·4 (44·2 to 62·6)	86·2 (74·2 to 100·8)
Other cardiovascular and circulatory diseases	153·3 (129·8 to 185·3)	265·9 (217·1 to 317
Chronic respiratory diseases	**2885·3 (2366·5 to 3234·0)**	**4163·7 (3612·7 to 5**
Chronic obstructive pulmonary disease	2404·0 (1975·9 to 2700·6)	3426·1 (2955·2 to 4
Pneumoconiosis	15·8 (12·3 to 22·4)	18·7 (14·1 to 24·9)
Silicosis	8·7 (6·4 to 13·1)	11·0 (7·7 to 15·2)
Asbestosis	1·8 (1·5 to 2·4)	3·0 (2·4 to 4·0)
Coal worker pneumoconiosis	3·3 (2·5 to 5·1)	2·3 (1·7 to 3·1)
Other pneumoconiosis	1·9 (1·4 to 2·9)	2·4 (1·6 to 3·6)
Asthma	342·7 (221·6 to 503·5)	441·9 (305·8 to 665
Interstitial lung disease and pulmonary sarcoidosis	76·3 (62·5 to 111·0)	213·2 (176·1 to 261
Other chronic respiratory diseases	46·5 (34·7 to 66·7)	63·6 (46·6 to 90·1)
Digestive diseases	**2052·3 (1849·8 to 2264·3)**	**2416·3 (2151·6 to 2**
Cirrhosis and other chronic liver diseases	1127·0 (1003·3 to 1275·7)	1282·0 (1141·9 to 14
Chronic hepatitis B including cirrhosis	391·2 (332·4 to 464·5)	394·2 (324·6 to 464
Chronic hepatitis C including cirrhosis	283·6 (236·1 to 343·1)	334·5 (276·9 to 403
Cirrhosis due to alcohol use	251·9 (215·4 to 291·7)	308·8 (259·4 to 358
Non-alcoholic fatty liver disease including cirrhosis	52·3 (37·2 to 69·9)	89·8 (64·7 to 120·3)
Cirrhosis due to other causes	148·1 (120·2 to 186·3)	154·7 (118·8 to 200
Upper digestive system diseases	316·1 (265·6 to 370·2)	265·4 (220·0 to 328
Peptic ulcer disease	273·7 (230·0 to 325·8)	222·4 (180·2 to 281
Gastritis and duodenitis	42·3 (29·3 to 57·6)	42·9 (27·1 to 58·5)
Appendicitis	33·2 (21·4 to 47·8)	31·0 (21·9 to 43·5)
Paralytic ileus and intestinal obstruction	184·9 (152·8 to 221·6)	243·7 (201·6 to 286
Inguinal, femoral, and abdominal hernia	39·0 (30·0 to 51·9)	51·1 (38·9 to 67·4)
Inflammatory bowel disease	27·5 (22·9 to 32·9)	46·5 (40·4 to 53·1)
Ulcerative colitis	22·7 (18·5 to 27·6)	39·6 (34·1 to 45·8)
Crohn's disease	4·7 (3·8 to 5·8)	7·0 (5·9 to 8·5)
Vascular intestinal disorders	67·8 (61·3 to 73·8)	92·6 (82·4 to 102·5)
Gallbladder and biliary diseases	74·1 (61·1 to 87·8)	146·7 (121·8 to 171
Pancreatitis	88·7 (75·8 to 110·6)	124·5 (107·8 to 147
Other digestive diseases	94·1 (77·9 to 117·0)	132·9 (113·6 to 154
Neurological disorders	**1316·1 (623·0 to 2702·2)**	**3024·3 (1357·2 to 6**
Alzheimer's disease and other dementias	934·3 (230·7 to 2338·9)	2214·6 (549·4 to 53
Parkinson's disease	190·7 (173·2 to 206·8)	427·1 (379·1 to 469
Idiopathic epilepsy	118·7 (90·3 to 150·2)	159·4 (125·4 to 202
Multiple sclerosis	12·2 (11·2 to 13·4)	19·1 (17·4 to 21·7)
Motor neuron disease	22·6 (20·2 to 24·7)	44·6 (40·8 to 50·0)
Other neurological disorders	37·7 (33·9 to 41·5)	159·5 (137·4 to 180
Mental disorders	**0·2 (0·1 to 0·3)**	**0·2 (0·1 to 0·4)**
Eating disorders	0·2 (0·1 to 0·3)	0·2 (0·1 to 0·4)
Anorexia nervosa	0·2 (0·1 to 0·3)	0·2 (0·1 to 0·4)
Substance use disorders	**254·5 (234·9 to 279·7)**	**344·1 (316·6 to 374**
Alcohol use disorders	170·2 (155·8 to 190·1)	171·6 (154·5 to 197
Drug use disorders	84·3 (73·2 to 96·7)	172·5 (149·2 to 198
Opioid use disorders	57·0 (49·1 to 65·6)	125·9 (108·2 to 144
Cocaine use disorders	6·4 (4·9 to 8·3)	17·6 (13·9 to 21·7)
Amphetamine use disorders	5·9 (3·9 to 9·6)	12·0 (9·6 to 15·0)
Other drug use disorders	15·0 (11·6 to 20·1)	17·0 (14·7 to 20·6)
Diabetes and kidney diseases	**1719·5 (1560·3 to 1893·2)**	**3535·6 (3142·0 to 3**
Diabetes mellitus	911·2 (784·3 to 1036·0)	2004·5 (1693·2 to 2
Type 1 diabetes mellitus	48·4 (40·6 to 60·3)	54·4 (42·3 to 71·6)
Type 2 diabetes mellitus	862·8 (740·0 to 985·7)	1950·1 (1646·3 to 2
Chronic kidney disease	796·0 (701·6 to 902·0)	1520·1 (1331·1 to 1
Chronic kidney disease due to type 1 diabetes mellitus	45·0 (35·2 to 57·9)	76·5 (56·4 to 98·4)
Chronic kidney disease due to type 2 diabetes mellitus	159·1 (123·4 to 194·4)	343·2 (271·1 to 414
Chronic kidney disease due to hypertension	189·4 (153·9 to 229·6)	442·3 (358·5 to 530
Chronic kidney disease due to glomerulonephritis	123·3 (101·2 to 145·5)	193·9 (161·9 to 228
Chronic kidney disease due to other and unspecified causes	279·2 (237·8 to 321·8)	464·1 (387·6 to 545
Acute glomerulonephritis	12·3 (7·9 to 18·4)	11·1 (7·5 to 14·7)
Skin and subcutaneous diseases	**66·0 (52·4 to 83·7)**	**161·1 (130·9 to 196**
Bacterial skin diseases	40·1 (28·2 to 52·9)	101·7 (82·5 to 128·7
Cellulitis	15·8 (10·4 to 23·6)	39·7 (28·9 to 55·5)
Pyoderma	24·2 (17·1 to 34·4)	62·0 (49·0 to 79·0)
Decubitus ulcer	21·2 (16·3 to 27·8)	48·1 (36·3 to 62·8)
Other skin and subcutaneous diseases	4·7 (2·8 to 8·3)	11·3 (6·9 to 17·6)
Musculoskeletal disorders	**80·2 (66·5 to 92·9)**	**131·7 (107·9 to 154**
Rheumatoid arthritis	28·8 (20·7 to 36·4)	45·9 (32·7 to 57·9)
Other musculoskeletal disorders	51·4 (42·1 to 60·6)	85·9 (67·9 to 103·6)
Other non-communicable diseases	**1121·0 (992·9 to 1256·6)**	**1338·5 (1191·5 to 1**
Congenital birth defects	659·3 (547·7 to 792·5)	562·7 (441·3 to 709
Neural tube defects	65·3 (39·4 to 107·4)	41·7 (24·6 to 69·1)
Congenital heart anomalies	386·7 (307·5 to 471·0)	301·2 (233·4 to 390
Orofacial clefts	11·5 (4·1 to 30·6)	3·3 (0·7 to 10·1)
Down syndrome	23·9 (15·4 to 34·4)	29·3 (17·8 to 43·7)
Other chromosomal abnormalities	14·9 (10·1 to 22·1)	22·9 (15·4 to 36·0)
Congenital musculoskeletal and limb anomalies	10·0 (6·6 to 16·3)	9·7 (6·2 to 15·0)
Urogenital congenital anomalies	12·9 (7·9 to 21·7)	16·9 (9·3 to 30·8)
Digestive congenital anomalies	63·1 (40·9 to 97·6)	67·0 (42·0 to 97·1)
Other congenital birth defects	71·1 (42·4 to 125·4)	70·9 (37·7 to 132·9)
Urinary diseases and male infertility	180·8 (162·1 to 199·0)	396·2 (357·0 to 433
Urinary tract infections and interstitial nephritis	122·2 (106·7 to 137·0)	288·5 (258·0 to 318
Urolithiasis	14·6 (12·1 to 17·2)	24·9 (21·0 to 30·0)
Other urinary diseases	44·0 (33·4 to 58·5)	82·8 (65·2 to 102·2)
Gynaecological diseases	5·3 (3·4 to 8·7)	14·1 (7·1 to 25·2)
Uterine fibroids	1·6 (0·9 to 2·9)	4·0 (2·0 to 7·4)
Endometriosis	0·0 (0·0 to 0·1)	0·2 (0·0 to 0·5)
Genital prolapse	0·6 (0·3 to 1·1)	1·6 (0·8 to 3·5)
Other gynaecological diseases	3·1 (1·7 to 5·2)	8·3 (3·9 to 15·2)
Haemoglobinopathies and haemolytic anaemias	121·9 (88·9 to 183·0)	132·4 (86·7 to 218·0
Thalassaemias	17·9 (13·9 to 24·6)	13·5 (9·1 to 19·6)
Sickle cell disorders	45·6 (26·5 to 85·7)	54·0 (27·0 to 110·5)
G6PD deficiency	12·6 (8·5 to 18·3)	12·9 (7·6 to 21·1)
Other haemoglobinopathies and haemolytic anaemias	45·9 (36·7 to 60·3)	51·9 (39·7 to 71·8)
Endocrine, metabolic, blood, and immune disorders	96·9 (84·3 to 108·9)	207·6 (179·5 to 233
Thyroid diseases	19·6 (14·9 to 25·2)	32·4 (25·6 to 41·1)
Other endocrine, metabolic, blood, and immune disorders	77·3 (67·1 to 86·5)	175·2 (153·3 to 197
Sudden infant death syndrome	56·7 (36·3 to 91·0)	25·6 (15·8 to 40·1)
Injuries	**4561·0 (4198·4 to 4847·2)**	**4874·5 (4365·5 to 5**
Transport injuries	**1370·0 (1150·1 to 1577·2)**	**1425·8 (1126·9 to 1**
Road injuries	1285·3 (1073·5 to 1485·5)	1343·7 (1044·8 to 1
Pedestrian road injuries	517·6 (403·1 to 627·3)	401·4 (294·8 to 531
Cyclist road injuries	62·3 (47·0 to 82·2)	89·1 (61·6 to 127·5)
Motorcyclist road injuries	216·7 (161·2 to 286·9)	319·3 (216·6 to 416
Motor vehicle road injuries	475·6 (394·0 to 598·0)	518·6 (416·7 to 648
Other road injuries	13·1 (8·7 to 18·9)	15·4 (10·0 to 22·6)
Other transport injuries	84·7 (60·9 to 110·7)	82·0 (55·8 to 112·2)
Unintentional injuries	**1791·4 (1590·1 to 1985·5)**	**2078·4 (1786·5 to 2**
Falls	487·5 (426·9 to 562·8)	857·4 (723·5 to 100
Drowning	432·5 (372·3 to 512·2)	291·1 (234·0 to 362
Fire, heat, and hot substances	134·6 (103·6 to 173·8)	150·7 (102·9 to 205
Poisonings	79·0 (65·0 to 99·8)	69·9 (54·4 to 91·7)
Poisoning by carbon monoxide	44·0 (37·4 to 52·6)	31·0 (24·4 to 39·9)
Poisoning by other means	35·0 (22·7 to 52·1)	38·9 (24·5 to 55·8)
Exposure to mechanical forces	127·9 (98·2 to 170·3)	104·1 (74·0 to 148·7
Unintentional firearm injuries	23·2 (13·8 to 37·9)	16·2 (10·0 to 25·8)
Other exposure to mechanical forces	104·7 (80·0 to 141·2)	87·9 (60·6 to 127·5)
Adverse effects of medical treatment	99·3 (77·6 to 127·7)	102·5 (85·3 to 128·6
Animal contact	102·2 (61·8 to 150·3)	103·1 (61·8 to 148·7
Venomous animal contact	92·4 (55·5 to 137·8)	93·7 (56·3 to 137·1)
Non-venomous animal contact	9·8 (5·9 to 18·1)	9·5 (5·2 to 14·9)
Foreign body	102·9 (79·8 to 125·1)	119·8 (91·3 to 148·7
Pulmonary aspiration and foreign body in airway	99·3 (77·0 to 119·7)	117·6 (90·0 to 144·7
Foreign body in other body part	3·6 (1·5 to 5·6)	2·3 (1·3 to 3·8)
Electrocution	56·7 (29·5 to 88·4)	42·4 (22·0 to 57·4)
Environmental heat and cold exposure	56·3 (42·6 to 69·6)	83·4 (71·5 to 95·7)
Exposure to forces of nature	9·0 (8·2 to 9·9)	88·6 (80·7 to 97·4)
Other unintentional injuries	103·4 (59·7 to 176·9)	65·2 (39·7 to 104·3)
Self-harm and interpersonal violence	**1399·6 (1273·1 to 1514·3)**	**1370·4 (1252·7 to 1**
Self-harm	819·5 (707·3 to 905·0)	766·7 (675·7 to 857
Self-harm by firearm	67·3 (50·2 to 98·4)	66·6 (51·5 to 86·1)
Self-harm by other specified means	752·2 (642·7 to 839·1)	700·2 (609·7 to 787
Interpersonal violence	464·3 (420·2 to 539·4)	435·7 (384·1 to 507
Physical violence by firearm	162·0 (145·0 to 185·9)	185·6 (167·5 to 209
Physical violence by sharp object	116·7 (94·2 to 140·4)	89·9 (67·3 to 125·8)
Physical violence by other means	185·6 (155·8 to 236·6)	160·2 (125·3 to 204
Conflict and terrorism	108·7 (94·2 to 145·1)	159·1 (125·6 to 208
Police conflict and executions	7·1 (5·5 to 9·1)	8·9 (6·3 to 13·3)
Total cancers	**7002·9 (6615·8 to 7281·6)**	**10 443·4 (9608·6 to**
Total burden related to hepatitis B	**586·6 (522·9 to 656·6)**	**628·4 (537·2 to 707**
Total burden related to hepatitis C	**387·4 (339·4 to 445·3)**	**497·2 (426·4 to 56**
Total burden related to non-alcoholic fatty liver disease	**72·7 (57·5 to 91·2)**	**131·5 (103·4 to 164**
Total cancers excluding non-melanoma skin cancer	**6972·7 (6588·6 to 7250·5)**	**10 379·5 (9553·2 to**

Values in parentheses are 95% uncertainty intervals. G6PD=glucose-6-phosphate dehydrogenase. NASH=non-alcoholic steatohepatitis. YLLs=years of life lost.

[*] Statistically significant percentage changes.

Causes of death

Figure 1 shows the global rankings of the leading Level 3 causes of age-standardised mortality rates over the period studied. From 1990 to 2023, ischaemic heart disease and stroke consistently ranked as the first and second leading causes, respectively—except in 2021, when COVID-19 temporarily ranked as the leading cause of age-standardised deaths. In 2021, the rankings of the leading five Level 3 causes, in descending order, were COVID-19, ischaemic heart disease, stroke, chronic obstructive pulmonary disease (COPD), and lower respiratory infections. In 2023, COVID-19 dropped to the 20th leading cause of death, with ischaemic heart disease, stroke, COPD, lower respiratory infections, and neonatal disorders ranking as the leading five causes. Although the rankings of the leading two causes of death in 2023, ischaemic heart disease and stroke, were the same as they were in 1990, the age-standardised mortality rates for each have decreased: ischaemic heart disease decreased from 161·4 deaths (95% UI 146·9–172·6) per 100 000 population in 1990, to 99·8 deaths (89·9–108·4) per 100

000 in 2023; while stroke declined from 157·2 deaths (141·1–171·3) per 100 000 in 1990 to 75·9 deaths (67·8–83·5) per 100 000 in 2023. Other notable shifts in the rankings of leading causes of death have occurred over the past three decades. Four causes showed declines in age-standardised mortality rates between 1990 and 2023: diarrhoeal diseases (67·4 deaths [55·6–85·5] per 100 000 in 1990 to 14·2 deaths [10·6–19·2] per 100 000 in 2023), tuberculosis (42·6 deaths [32·4–54·2] per 100 000 in 1990 to 11·6 deaths [9·2–14·4] per 100 000 in 2023), stomach cancer (23·3 deaths [20·6–26·5] per 100 000 in 1990 to 10·3 deaths [8·8–11·9] per 100 000 in 2023), and measles (18·1 deaths [7·8–30·4] per 100 000 in 1990 to 2·2 deaths [0·9–3·9] per 100 000 in 2023). By contrast, some causes exhibited an increase in age-standardised mortality rates between 1990 and 2023, such as diabetes, chronic kidney disease, Alzheimer's disease and other dementias, and HIV/AIDS.

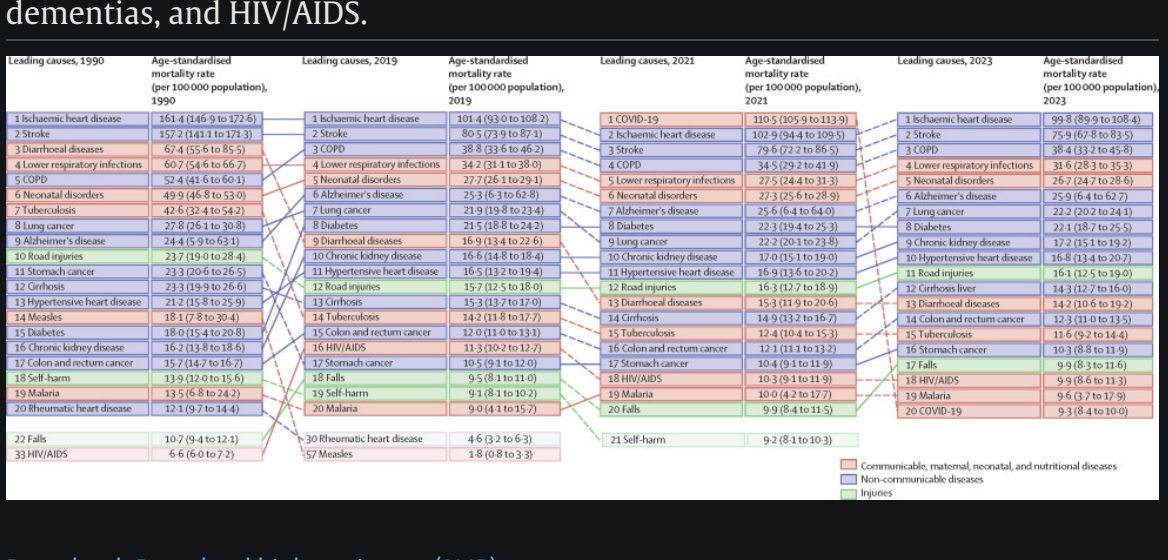


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Figure 1. Leading Level 3 causes of global deaths and age-standardised mortality rate per 100 000 population for all sexes combined, 1990, 2019, 2021, and 2023
The 20 leading causes of death are shown in descending order. Causes are connected by lines between time periods; solid lines represent an increase or lateral shift in rank and dashed lines represent decreases in rank. Alzheimer's disease=Alzheimer's disease and other dementias. Cirrhosis=cirrhosis and other chronic liver diseases. COPD=chronic obstructive pulmonary disease. Lung cancer=tracheal, bronchus, and lung cancer.

The percentage change among leading Level 3 causes of death varied over the study period between females and males at the global level (figure 2). The age-standardised mortality rates for HIV/AIDS, urinary diseases, chronic kidney disease, and COPD increased more among females than males. Likewise, those for falls, asthma, and hypertensive heart disease decreased more among males than females. Deaths from conflict and terrorism were particularly disparate by sex and location (appendix 2 table S18). Between 1995 and 2023, 51·3% (95% UI 45·9–53·9) of female deaths due to conflict and terrorism occurred in north Africa and the Middle East, despite only 7·2% of the global female population residing in this region. Eastern Europe contributed 7·5% (6·3–8·3) of female deaths from this cause, despite having only 3·4% of the global female population. Over the same time period, 52·0% (46·3–54·7) of male deaths due to conflict and terrorism occurred in north Africa and the Middle East, which accounts for only 7·7% of the global male population, and 8·0% (6·6–9·1) in eastern Europe, with only 2·9% of the global male population (appendix 2 table S18). In 2023, Palestine had the highest age-standardised mortality rate due to conflict and terrorism of any country in the world (385·8 deaths [351·3–424·1] per 100 000 population), more than five times that of the second-leading country, Ukraine (70·1 deaths [67·2–73·2] per 100 000). Sudan ranks as the third highest country in terms of age-standardised mortality rates for conflict and terrorism, while Russia and Burkina Faso follow as the fourth and fifth leading countries, in 2023 (appendix 2 table S7).

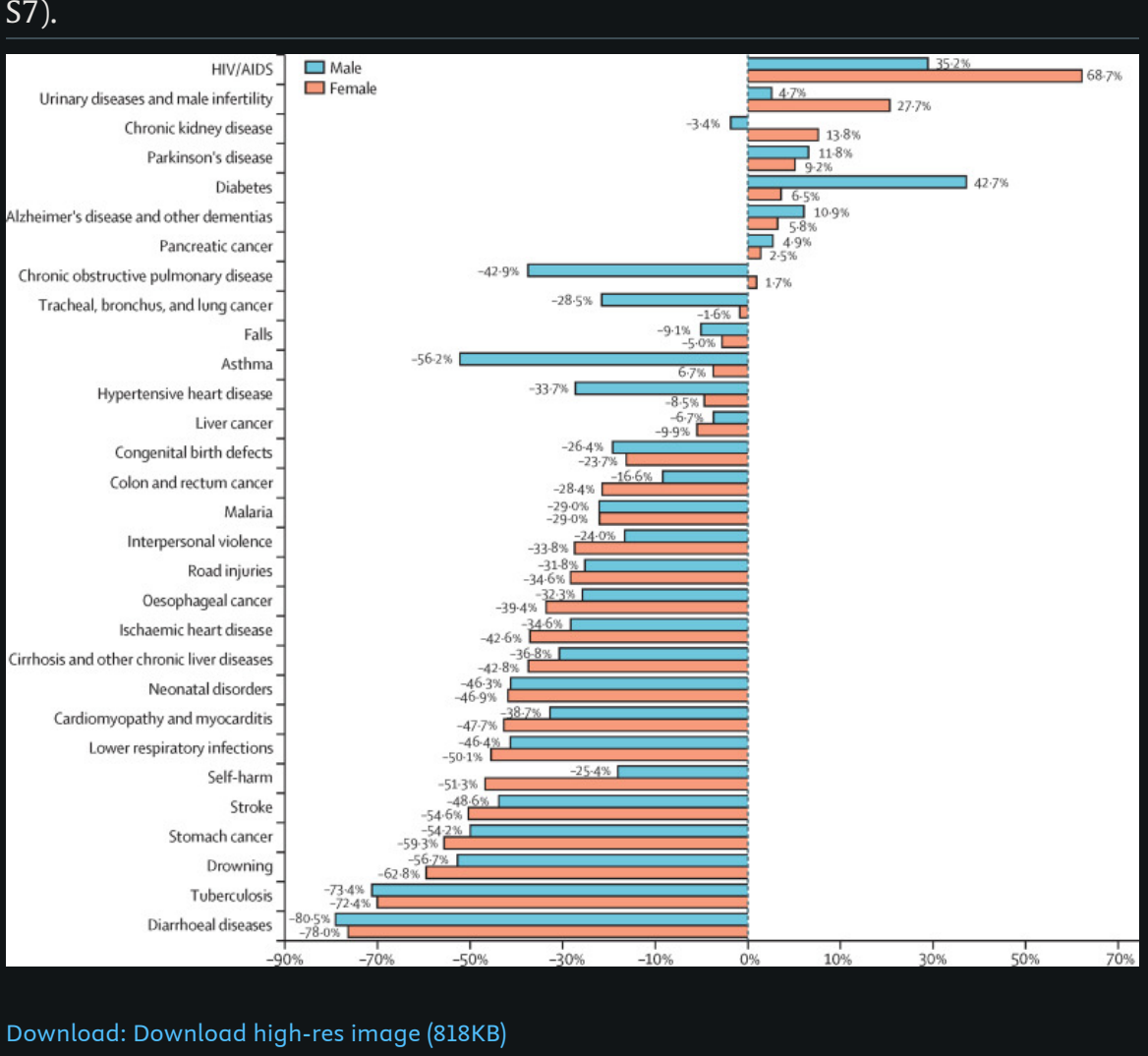

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Figure 2. Percentage change in global age-standardised mortality rate from 1990 to 2023 among the leading 30 Level 3 causes of death, for males and females
Figure shows the top 30 causes according to their global age-standardised mortality rate, sorted by percentage change from 1990 to 2023 in females, in descending order. COVID-19 and causes affecting only one sex (ie, cervical cancer) were omitted.

Causes of YLLs

Over the study period, neonatal disorders remained the leading Level 3 cause of global YLLs, despite a decrease in age-standardised YLL rates, from 4487·8 YLLs (95% UI 4212·3–4761·6) per 100 000 population in 1990 to 2398·9 YLLs (2219·7–2575·4) per 100 000 in 2023, representing a decrease of 46·5% (−51·2 to −41·6; appendix 2 table S13). In 2021, however, COVID-19 temporarily surpassed neonatal disorders as the leading cause of global age-standardised YLLs, before dropping to the 25th position in 2023 (appendix figure S2). Since the year 2000, there has been a reduction of 37·6% (32·6–42·6) in total YLLs due to neonatal disorders, from 235 000 in 2000 to 146 000 in 2023 (table 1). In 1990, the total YLLs for vaccine-preventable diseases—including diphtheria, pertussis, tetanus, measles, varicella and herpes zoster, yellow fever, rabies, liver cancer due to hepatitis B, cervical cancer, chronic hepatitis B with cirrhosis, and acute hepatitis B—amounted to a sum of 178 million (95% UI 122–239) years (appendix 2 table S4). By 2023, this number had decreased by 66·5% (57·4–71·9) to 59·8 million (49·6–72·1) years. Similarly, for other preventable diseases that went through major international cooperation and large-scale interventions between 1990 and 2023, total YLLs also showed a decline. In 1990, total YLLs for these diseases were 1·07 billion (1·01–1·12) years; by 2023, they had decreased by 51·6% (48·8–54·4) to 516 million (488–543) years (appendix 2 table S4).

COVID-19

Between 2020 and 2023, COVID-19 had an immense global impact, resulting in 18·0 million (95% UI 17·2–18·7) total deaths (appendix 2 table S5). Among these, 10·5 million (10·1–10·9) deaths were in males, and 7·53 million (6·98–7·96) deaths were in females. During 2020, the first year of the pandemic, there were 5·47 million (5·22–5·65) COVID-19 deaths. The highest number of COVID-19 deaths was recorded in 2021 (9·42 million [9·04–9·70] deaths) followed by 2·36 million (2·19–2·47) deaths in 2022, and the lowest number of deaths occurred in 2023 (797 000 deaths [723 000–857 000]). The age group most affected by COVID-19 varied over the years: the highest numbers of deaths occurred in the 70–74 years age group in 2020 and 2021, but in the 80–84 years age group in 2022, and in the 85–89 years age group in 2023 (appendix 2 figure S3). From 2020 to 2023, the five countries with the highest numbers of total COVID-19 deaths (in descending order) were India (3·08 million [2·92–3·28] deaths), the USA (1·21 million [1·10–1·27] deaths), Russia (1·06 million [1·00–1·09] deaths), Indonesia (849 000 deaths [780 000–920 000]), and Brazil (795 000 deaths [748 000–821 000]; appendix 2 table S5). In mortality rates, the countries with the highest burden (in descending order) were Tunisia (245·4 deaths [227·9–256·9] per 100 000), Bolivia (229·0 deaths [215·5–241·4] per 100 000), Peru (220·6 deaths [213·4–228·5] per 100 000), Montenegro (215·2 deaths [197·9–229·3] per 100 000), and Moldova (207·3 deaths [199·4–212·7] per 100 000; appendix 2 table S5).

Changes in age-specific mortality rate from 2000 to 2023

Globally, between 2000 and 2023, all 5-year age groups from 10 years to 70 years showed decreases in age-specific mortality rates, ranging from −27·1% to −35·0% (table 2; appendix 2 table S19). However, this trend varied by country, region, sex, and cause of death. Among the 21 GBD regions, only four—high-income Asia Pacific, central Asia, east Asia, and south AsiA&Mdash;consistently showed decreases in death rates across all 5-year age groups by sex and Level 1 causes from 2000 to 2023. By contrast, the Caribbean region showed increases in death rates for 40 distinct Level 1 cause-sex-age groups; the largest of these increases was observed in females aged 30–34 years, in whom deaths from NCDs increased by 52·6%, and in females aged 25–29 years, in whom NCD deaths rose by 48·9%. Similarly, in high-income North America, there was an increase in death rates across 37 distinct Level 1 cause-sex-age groups. The most notable rise was in males aged 25–29 years, where NCD-related deaths rose by 130·5%, while injury-related deaths increased by 26·4% in males aged 60–64 years. Globally, we found the increase in NCD deaths was primarily due to drug use disorders, while the rise in injury-related deaths was linked to intentional injuries. Other notable injury-related increases around the globe include the conflicts in Palestine and Ukraine, as well as specific natural disasters such as the 2023 earthquake in Türkiye and the 2022–23 heatwaves in Europe (appendix 2 table S19).

Table 2. Percentage change in age-specific mortality rate between 2000 and 2023 for Level 1 causes

	Age 10–14 years	Age 15–19 years	Age 20–24 years	Age 25–29 years	Age 30–3
Communicable, maternal, neonatal, and nutritional diseases					
Caribbean					
Male	−32·0%	−22·6%	−33·9%	−46·5%	−50·7%
Female	−23·6%	−16·3%	−24·6%	−38·1%	−36·2%
High-income North America					
Male	−11·1%	−22·5%	−20·9%	−38·1%	−54·0%
Female	−23·7%	−21·6%	−18·5%	−20·0%	−24·4%
Central sub-Saharan Africa					
Male	−59·9%	−66·8%	−67·2%	−65·6%	−64·5%
Female	−56·6%	−52·3%	−56·3%	−60·4%	−61·1%
Oceania					
Male	−34·2%	−30·9%	−18·5%	−18·6%	−20·0%
Female	−32·1%	−27·4%	−13·1%	−5·1%	−0·7%
Western sub-Saharan Africa					
Male	−48·0%	−43·3%	−46·4%	−53·9%	−59·6%
Female	−44·8%	−48·6%	−56·7%	−58·2%	−56·6%
Southern sub-Saharan Africa					
Male	−9·0%	37·4%	−42·0%	−72·8%	−75·8%
Female	−16·9%	−37·9%	−71·6%	−81·3%	−75·6%
Southern Latin America					
Male	−19·0%	−6·0%	−17·0%	−42·1%	−49·4%
Female	−20·8%	−13·5%	−17·5%	−25·0%	−21·5%
Tropical Latin America					
Male	−20·8%	−8·7%	−10·6%	−27·0%	−42·0%
Female	−28·0%	−21·0%	−24·1%	−32·7%	−37·1%
Eastern sub-Saharan Africa					
Male	−63·2%	−61·4%	−61·9%	−71·3%	−76·1%
Female	−64·5%	−64·6%	−72·2%	−78·8%	−79·0%
North Africa and Middle East					
Male	−52·1%	−30·7%	−27·9%	−35·2%	−37·5%
Female	−55·9%	−41·5%	−44·4%	−45·0%	−44·3%
Eastern Europe					
Male	−40·5%	−57·4%	−78·1%	−62·0%	−35·7%
Female	−33·3%	−41·3%	−44·9%	−2·3%	35·7%
Central Latin America					
Male	−33·7%	−22·7%	−30·0%	−31·5%	−30·5%
Female	−35·2%	−26·8%	−28·7%	−27·7%	−22·1%
Central Europe					
Male	−67·2%	−17·0%	−6·0%	−13·4%	−19·1%
Female	−63·5%	−26·0%	−24·9%	−22·4%	−25·8%
Western Europe					
Male	−46·9%	−51·5%	−56·4%	−68·3%	−73·7%
Female	−43·6%	−49·9%	−42·2%	−52·3%	−57·0%
Southeast Asia					
Male	−50·6%	−47·0%	−43·6%	−54·7%	−58·0%
Female	−49·9%	−52·5%	−50·6%	−53·1%	−51·4%
Andean Latin America					
Male	−39·1%	−39·2%	−39·2%	−40·5%	−40·0%
Female	−36·7%	−47·2%	−46·4%	−46·7%	−45·6%
Australasia					
Male	−29·1%	−50·8%	−44·1%	−66·6%	−62·6%
Female	−42·0%	−50·7%	−38·6%	−45·4%	−49·7%
South Asia					
Male	−74·2%	−71·2%	−71·8%	−73·1%	−70·3%
Female	−70·6%	−75·6%	−76·2%	−74·3%	−71·0%
High-income Asia Pacific					
Male	−54·1%	−59·7%	−50·2%	−49·5%	−58·8%
Female	−57·1%	−52·2%	−45·3%	−51·3%	−58·3%
East Asia					
Male	−60·6%	−56·6%	−38·2%	−22·9%	−24·3%
Female	−69·8%	−71·9%	−72·5%	−69·7%	−67·5%
Central Asia					
Male	−48·9%	−57·3%	−75·4%	−79·2%	−75·0%
Female	−48·5%	−53·6%	−65·4%	−68·0%	−62·3%
Non-communicable diseases					
Caribbean					
Male	−4·4%	2·1%	3·1%	14·9%	17·8%
Female	13·0%	8·0%	17·4%	48·9%	52·6%
High-income North America					
Male	−9·7%	29·5%	79·3%	130·5%	122·5%
Female	−9·5%	16·3%	50·5%	67·0%	63·3%
Central sub-Saharan Africa					
Male	−13·9%	−30·2%	−31·7%	−28·2%	−15·7%
Female	−4·7%	−6·6%	−9·5%	30·0%	56·5%
Oceania					
Male	−12·8%	−18·5%	−1·1%	16·1%	10·0%
Female	−13·4%	−17·7%	0·9%	28·0%	46·9%
Western sub-Saharan Africa					
Male	−18·3%	−19·6%	−22·3%	−19·7%	−11·2%
Female	−23·0%	−24·9%	−30·5%	1·2%	16·3%
Southern sub-Saharan Africa					
Male	−15·6%	−27·8%	−26·1%	−21·2%	−16·4%
Female	−21·8%	−13·5%	−5·1%	8·9%	23·9%
Southern Latin America					
Male	−23·9%	−14·4%	−6·9%	−10·1%	−20·6%
Female	−23·5%	−12·4%	−9·6%	−8·8%	−13·8%
Tropical Latin America					
Male	1·3%	12·9%	14·0%	−0·3%	−15·8%
Female	0·4%	2·9%	7·0%	3·2%	−6·9%
Eastern sub-Saharan Africa					
Male	−20·1%	−26·8%	−30·6%	−24·9%	−17·1%
Female	−27·9%	−32·0%	−30·5%	−0·4%	12·6%
North Africa and Middle East					
Male	−28·3%	−16·5%	−7·0%	−5·6%	−12·2%
Female	−33·7%	−25·8%	−11·9%	−6·2%	−2·7%
Eastern Europe					
Male	−28·8%	−39·7%	−62·2%	−49·1%	−30·6%
Female	−32·2%	−37·8%	−39·7%	−26·8%	−16·5%
Central Latin America					
Male	0·0%	7·6%	5·4%	5·5%	3·7%
Female	2·9%	1·5%	−0·5%	−1·4%	3·5%
Central Europe					
Male	−45·4%	−31·8%	−23·1%	−19·1%	−20·2%
Female	−30·7%	−25·4%	−22·4%	−22·0%	−22·4%
Western Europe					
Male	−30·0%	−33·7%	−31·5%	−29·4%	−24·1%
Female	−20·4%	−22·0%	−19·0%	−21·8%	−18·6%
Southeast Asia					
Male	−23·7%	−22·7%	−13·8%	−6·8%	−10·5%
Female	−14·3%	−10·2%	−5·1%	0·1%	7·4%
Andean Latin America					
Male	−6·5%	−5·1%	−11·0%	−5·8%	−10·2%
Female	14·0%	−5·6%	−7·0%	−5·2%	−3·2%
Australasia					
Male	−35·5%	−43·8%	−42·7%	−46·5%	−32·2%
Female	−33·4%	−40·1%	−32·6%	−35·6%	−33·2%
South Asia					
Male	−39·5%	−47·3%	−44·6%	−41·5%	−33·4%
Female	−39·7%	−44·6%	−35·1%	−26·9%	−11·7%
High-income Asia Pacific					
Male	−32·9%	−35·2%	−33·1%	−37·3%	−45·0%
Female	−28·0%	−25·4%	−23·4%	−30·9%	−37·0%
East Asia					
Male	−47·6%	−49·4%	−57·8%	−54·2%	−52·2%
Female	−49·5%	−58·1%	−65·5%	−66·7%	−64·4%
Central Asia					
Male	−19·6%	−32·9%	−46·0%	−50·4%	−44·6%
Female	−16·0%	−24·9%	−38·2%	−41·4%	−36·0%
Injuries					
Caribbean					
Male	−5·2%	31·0%	26·4%	35·3%	36·8%
Female	10·1%	11·6%	24·2%	45·7%	42·4%
High-income North America					
Male	−23·3%	−21·9%	−19·2%	1·6%	20·0%
Female	−18·9%	−31·0%	−7·5%	6·3%	14·6%
Central sub-Saharan Africa					
Male	−4·9%	−23·1%	−23·2%	−15·9%	2·8%
Female	2·3%	5·1%	4·3%	51·8%	66·9%
Oceania					
Male	−19·8%	−24·8%	−6·3%	8·5%	1·0%
Female	−12·6%	−15·9%	9·4%	30·6%	45·8%
Western sub-Saharan Africa					
Male	−19·2%	−19·4%	−23·4%	−21·6%	−12·9%
Female	−27·6%	−33·0%	−30·2%	−1·7%	7·1%
Southern sub-Saharan Africa					
Male	−38·2%	−37·0%	−26·2%	−20·4%	−16·5%
Female	−42·4%	−16·7%	−11·0%	−2·2%	0·4%
Southern Latin America					
Male	−53·0%	−31·2%	−22·0%	−21·1%	−24·3%
Female	−45·7%	−26·3%	−15·1%	−15·0%	−22·8%
Tropical Latin America					
Male	−47·2%	−14·1%	−6·2%	−6·5%	−11·5%
Female	−38·7%	−14·9%	2·8%	5·2%	−1·9%
Eastern sub-Saharan Africa					
Male	−27·8%	−63·3%	−55·1%	−35·1%	−24·8%
Female	−47·2%	−49·0%	−50·0%	−20·7%	−17·5%
North Africa and Middle East					
Male	−9·1%	−0·1%	15·1%	−5·7%	−14·0%
Female	30·4%	24·2%	55·2%	45·4%	50·3%
Eastern Europe					
Male	−48·7%	9·2%	−2·3%	−11·4%	−21·6%
Female	1·1%	−30·6%	−37·0%	−45·1%	−47·6%
Central Latin America					
Male	−35·2%	−24·5%	−21·5%	−13·7%	−7·8%
Female	−19·9%	−12·2%	−4·6%	−0·5%	3·4%
Central Europe					
Male	−64·9%	−51·2%	−45·5%	−41·2%	−38·6%
Female	−52·9%	−43·3%	−43·0%	−40·2%	−38·1%
Western Europe					
Male	−57·2%	−51·2%	−47·6%	−41·5%	−31·8%
Female	−51·4%	−49·7%	−38·7%	−34·5%	−29·0%
Southeast Asia					
Male	−43·0%	−36·2%	−35·1%	−34·7%	−32·6%
Female	−39·2%	−27·6%	−22·5%	−21·9%	−15·2%
Andean Latin America					
Male	−30·6%	−12·1%	1·6%	11·2%	1·4%
Female	−2·4%	−4·2%	0·5%	4·8%	0·5%
Australasia					
Male	−51·4%	−53·3%	−46·9%	−48·2%	−40·4%
Female	−47·0%	−45·1%	−29·2%	−34·3%	−35·7%
South Asia					
Male	−46·2%	−42·2%	−33·0%	−24·3%	−15·7%
Female	−47·3%	−48·5%	−46·3%	−36·4%	−28·3%
High-income Asia Pacific					
Male	−51·6%	−48·3%	−35·5%	−32·5%	−38·9%
Female	−35·5%	−13·8%	−1·8%	−6·1%	−17·1%
East Asia					
Male	−67·3%	−68·8%	−73·6%	−70·7%	−70·5%
Female	−66·1%	−71·2%	−78·3%	−79·7%	−78·6%
Central Asia					
Male	−36·8%	−39·4%	−50·7%	−54·3%	−52·2%
Female	−34·1%	−19·6%	−40·4%	−49·3%	−48·6%

Regions are ordered by the total number of cause-age combinations that showed an increase across all three Level 1 causes.

Mean age at death

The mean age at death for all causes varied by sex and location (appendix 2 tables S15, S16, S17). The global mean age at death increased from 46·8 years (95% UI 46·6–47·0) in 1990 to 63·4 years (63·1–63·7) in 2023, for all sexes combined. For males, the mean age at death in 1990 was 45·4 years (45·1–45·7) and increased to 61·2 years (60·7–61·6) in 2023. For females, it was 48·5 years (48·1–48·8) in 1990 and 65·9 years (65·5–66·3) in 2023. The highest mean age at death observed in 2023 was found in the high-income super-region. Within this super-region, mean age at death for females reached 80·9 years (80·9–81·0), and was even higher in the high-income Asia Pacific region (85·1 years [85·1–85·2]), with Japan having the highest mean among all countries globally (86·0 years [86·0–86·1]). For males in the high-income super-region, the mean age at death was 74·8 years (74·8–74·9). In high-income Asia-Pacific, it was 78·6 years (78·5–78·6), and Japan also recorded the highest male mean age at death at 79·8 years (79·8–79·8).

At the other end of the spectrum, the lowest mean age at death in 2023 occurred in sub-Saharan Africa, where females had a mean age at death of 38·0 years (95% UI 37·5–38·4; appendix 2 table S16). For males, it was 35·6 years (35·2–35·9; appendix 2 table S17). Within this super-region, western sub-Saharan Africa had a mean age at death of 33·2 years (32·5–34·0) for females and 31·9 years (31·2–32·6) for males. Niger recorded the lowest mean age at death, with 21·5 years (20·3–22·8) for females and 21·8 years (20·6–23·2) for males.

Mean age at death by cause

In 2023, the gap between the observed and the expected mean age at death varied across causes, locations, and sexes (figure 3; appendix 2 figure S4, appendix 2 table S6). For the global leading cause of death, ischaemic heart disease, females in Switzerland died at the highest mean age of 88·4 years (95% UI 87·8–88·8), which is 6·8 years (6·1–7·5) higher than the expected age of 81·6 (80·7–82·4). By contrast, females in South Sudan died from the same cause at the lowest mean age of 61·2 years (58·9–63·5), which is 7·3 years (4·8–10·1) below the expected age of 68·5 years (67·1–70·2). For tracheal, bronchus, and lung cancer—the sixth-leading cause of death—females in Japan died at the highest mean age of 82·8 years (81·3–83·5), which is 4·2 years (3·6–4·6) later than the expected age of 78·6 (77·4–79·4). However, in Malawi, females died from this cause at the lowest mean age of 55·6 years (53·9–57·9), which is 12·8 years (10·4–14·6) earlier than the expected age of 68·4 (67·2–69·3). For the ninth-leading cause of death, chronic kidney disease, females in Spain died at the second highest mean age of 89·4 years (89·0–89·7), 9·8 years (8·8–11·1) later than the expected age of 79·6 (78·0–80·8). Meanwhile, in Angola, the same cause resulted in a mean age at death of just 46·1 years (42·9–50·3), which is 12·5 years (9·2–15·6) earlier than the expected age of 58·6 (56·1–61·2).

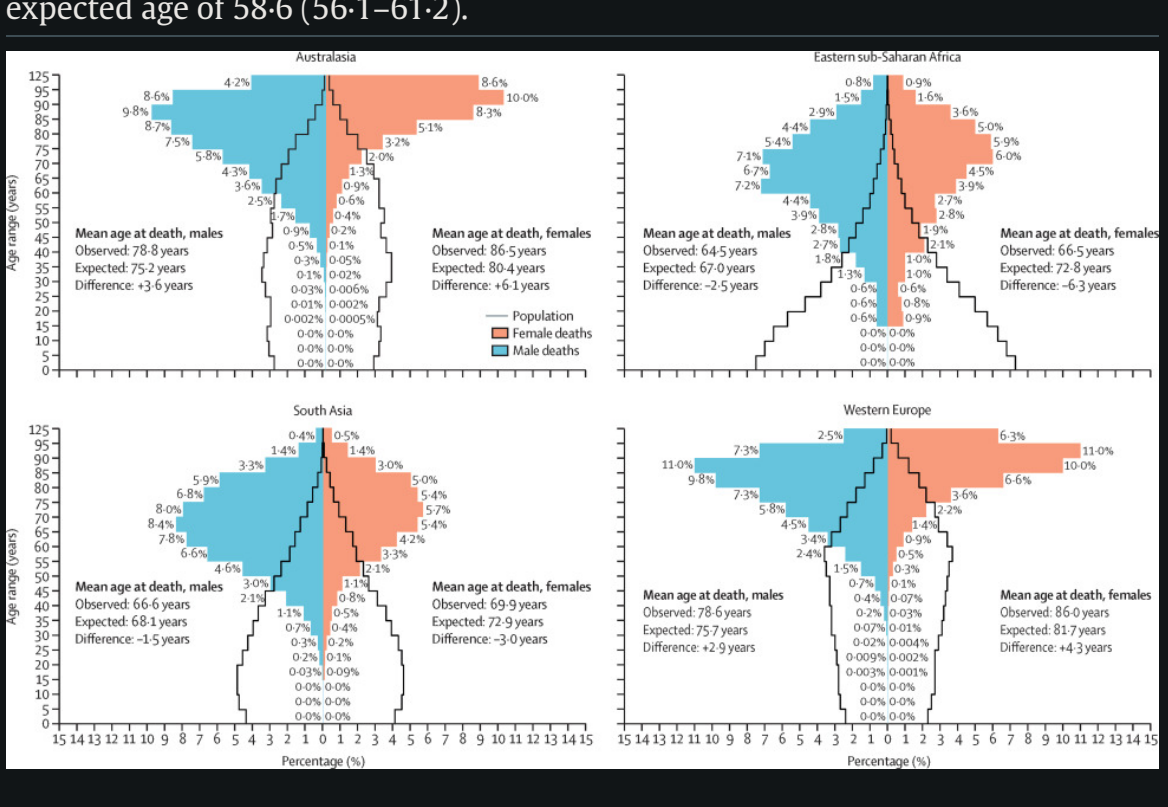

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Figure 3. Comparison of age at death for ischaemic heart disease between four regions for males and females
Graphs show the distribution of ischaemic heart disease deaths by age and sex within each region. Percentages represent the number of ischaemic heart disease deaths for a given age-sex group out of the total ischaemic heart disease deaths within a region (all ages and sexes combined), or the total number of individuals in a given age-sex group out of the total population in the region. The expected mean age at death is the result of calculating the mean age at death after applying the global mortality rate to a country's population by age and sex for a given cause; a positive difference indicates that the observed mean age at death is higher than expected.

The observed mean age at death in 2023 shows a moderate relationship ($r^2 \geq 0·50$) with SDI across 96 of 141 Level 3 causes of death (appendix 2 table S14). Where SDI explains greater than 50% of the variance seen in the mean age at death, the relationship is positive: as SDI improves, the observed mean age at death increases. After controlling for any relationship SDI has with population structure by comparing observed and expected mean ages and examining their correlation with SDI, the results remain varied by sex.

When comparing observed and expected mean ages for females, a total of 118 causes show a positive correlation with SDI (appendix 2 table S14). Some causes, such as self-harm, exhibit a negative correlation—meaning that females in higher SDI countries died at younger ages than expected from self-harm compared to those in lower SDI countries. All of the causes that had a negative correlation with SDI were considered weak relationships ($r^2 < 0·50$). For females, the difference between the observed and expected mean age at death in the following nine Level 3 causes showed a moderate positive correlation with SDI: ischaemic heart disease, stroke, breast cancer, pancreatic cancer, leukaemia, brain and central nervous system cancer, ovarian cancer, kidney cancer, and invasive non-typhoidal *Salmonella*.

When comparing observed and expected mean ages for males, a total of 110 causes show a positive correlation with SDI (appendix 2 table S14). Similar causes, such as drug use disorders, self-harm, and conflict and terrorism exhibited a negative correlation—indicating that males in higher SDI countries died at younger ages than expected from these causes compared to those in lower SDI countries. All negatively correlated causes were found to have weak relationships with SDI just as females did. For males, the difference between the observed and expected mean age at death in only six Level 3 causes showed a moderate correlation with SDI. These were leukaemia, brain and central nervous system cancer, other malignant neoplasms, other intestinal infectious diseases, kidney cancer, and invasive non-typhoidal *Salmonella*.

All-cause 70q0

Across every GBD super-region and region, 70q0 from all causes combined decreased for both males and females between 2000 and 2023 (appendix 2 table S11). However, there was variation in these percentage changes between regions. For males, 70q0 in the Caribbean decreased 2·2%, and in high-income North America it decreased 9·6%. Conversely, in high-income Asia Pacific, this decrease was 36·0%, and in east Asia it was 43·8%. For females, 70q0 in Oceania decreased 2·6% and in the Caribbean 6·0%, while in eastern Europe the decrease was 35·6% and in east Asia it was 58·2%.

National-level trends in 70q0 also varied (figure 4, appendix 2 table S11). For males, between 2000 and 2023, an increase in 70q0 occurred in six countries: Palestine (40·6%), Lebanon (19·4%), Guam (14·9%), Paraguay (14·6%), Dominican Republic (9·2%), and Venezuela (3·5%). For females, for the same period, 12 countries had an increase in 70q0: Libya (19·6%), Palestine (14·2%), Lebanon (13·7%), Venezuela (12·8%), Tonga (12·0%), Solomon Islands (7·8%), Samoa (6·6%), Guam (4·9%), Marshall Islands (4·7%), Paraguay (4·6%), Dominican Republic (1·2%), and Fiji (0·8%). For males, the primary cause of death driving these increases for Palestine and Lebanon was conflict and terrorism, while in Paraguay and Guam the primary driver was drug use disorders. Among females, the primary cause responsible for the increases remained conflict and terrorism for Palestine and Lebanon. Additionally, chronic kidney diseases contributed the most to the increases in Libya, and malaria the most to the increases in Venezuela (appendix increases in Libya, and malaria the most to the increases in Venezuela (appendix 2 table S6)

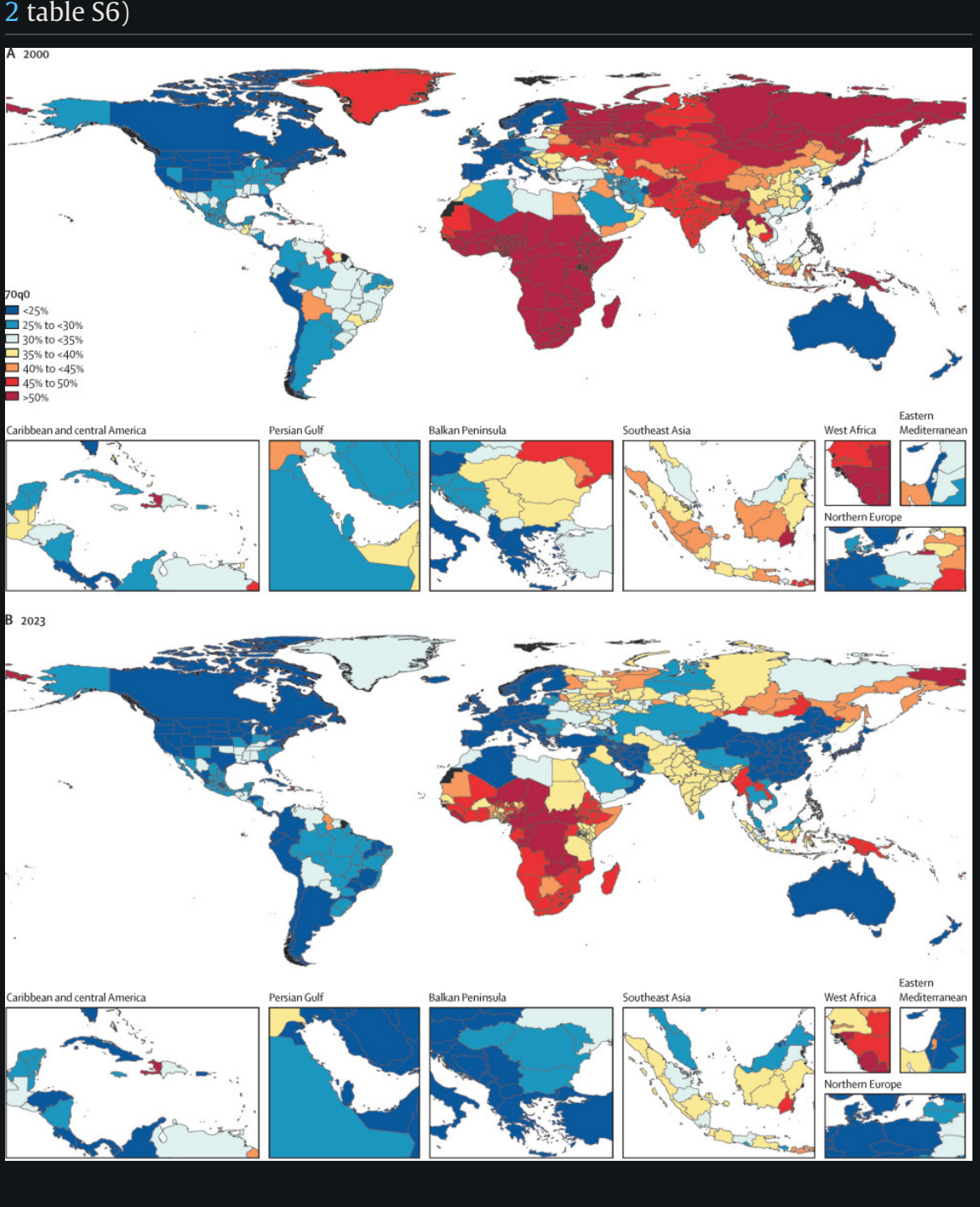

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Figure 4. 70q0 in males and females
(A) 2000. (B) 2023. 70q0=probability of death before age 70 years.

70q0 by cause and location

Among the top 50 global causes of death for females, the 15 causes with the largest amount of increase in 70q0 between 2000 and 2023 were interstitial lung disease and pulmonary sarcoidosis (+74·8% change; 70q0 in 2023 0·1%); drug use disorders (+68·0%; 0·1%); lip and oral cavity cancer (+54·7%; 0·1%); atrial fibrillation and flutter (+52·2%; <0·1%); Alzheimer's disease and other dementias (+46·2%; 0·2%); diabetes (+42·7%; 0·8%); hypertensive heart disease (+40·9%; 0·4%); pancreatic cancer (+37·7%; 0·2%); Parkinson's disease (+36·7%; <0·1%); ovarian cancer (+36·5%; 0·3%); other cardiovascular and circulatory diseases (+35·9%; 0·1%); breast cancer (+33·6%; 1·0%); endocrine, metabolic, blood, and immune disorders (+29·9%; 0·1%); chronic kidney disease (+27·7%; 0·6%); and non-rheumatic valvular heart disease (+26·2%; <0·1%; appendix 2 table S6).

Among the top 50 global causes for males, the 15 causes with the largest amount of increase in 70q0 between 2000 and 2023 were diabetes (+75·6% change; 70q0 in 2023 1·0%); drug use disorders (+56·8%; 0·2%); atrial fibrillation and flutter (+51·1%; <0·1%); interstitial lung disease and pulmonary sarcoidosis (+48·2%; 0·1%); Alzheimer's disease and other dementias (+45·9%; 0·1%); Parkinson's disease (+44·4%; 0·1%); endocrine, metabolic, blood, and immune disorders (+41·3%; 0·1%); pancreatic cancer (+34·9%; 0·3%); lip and oral cavity cancer (+29·2%; 0·2%); prostate cancer (+24·7%; 0·2%); non-rheumatic valvular heart disease (+23·9%; <0·1%); chronic kidney disease (+18·6%; 0·7%); colon and rectum cancer (+18·0%; 0·6%); urinary diseases and male infertility (+18·0%; 0·1%); and other cardiovascular and circulatory diseases (+15·1%; 0·1%; appendix 2 table S6).

Among the top 50 causes for which a global increase in 70q0 occurred, we observed several notable declines at the super-region level between 2000 and 2023 (appendix 2 table S6). For example, 70q0 due to drug use disorders increased by 56·8% in males and 68·0% in females globally, yet decreased in southeast Asia, east Asia, and Oceania by 73·9% for males and 81·5% for females. For diabetes, 70q0 increased by 75·6% for males and 42·7% for females globally, but decreased by 9·0% for males and 34·6% for females in the high-income super-region. For ovarian cancer, there was a 36·5% increase in 70q0 in females globally, but a 25·5% decrease in the high-income super-region. For chronic kidney disease, 70q0 increased by 18·6% for males and 27·7% for females globally, but in central Europe, eastern Europe, and central Asia, it decreased by 10·4% for males and 15·3% for females.

Alternatively, some causes of death showed increased national 70q0 where there has otherwise been global progress to reduce 70q0. For example, a 47·4% decrease in 70q0 due to lower respiratory infections occurred among males globally, yet there were substantial increases in countries such as Poland (98·8%), Thailand (82·7%), and Argentina (66·0%). Similarly, for females, there was a 51·2% decrease in 70q0 due to lower respiratory infections globally, but notable increases in a number of countries, including Argentina (101·5%), Poland (48·2%), and Thailand (17·1%). Road injuries are another example: a 21·1% decline in 70q0 among males occurred globally, despite increases in many countries, most notably in Sierra Leone (259·8%), Uganda (180·5%), and Malawi (155·0%). For road injuries among females, there was a 20·3% decrease in 70q0 globally, with notable increases in the Democratic Republic of the Congo (158·5%), Sierra Leone (122·1%), and Pakistan (121·7%). Additionally, stroke showed global decreases in 70q0 of 15·6% among males and 20·7% among females, but increases in many countries for males (eg, Rwanda [97·0%], Burundi [85·7%], and Ethiopia [75·6%]) and females (eg, Ethiopia [128·4%], Zimbabwe [106·7%], and South Sudan [106·7%]).

Joint examination of 70q0 and mean age at death in super-regions

From 2000 to 2023, notable variation was observed between sexes when investigating the 70q0 and mean age at death metrics for the top 20 causes of death by super-region (figures 5, 6). For females in sub-Saharan Africa, several of the leading causes showed an increase in 70q0 and a mean age that was lower than expected: ischaemic heart disease (70q0 increased 81·1%, with mean age at death 3·4 years lower than expected); stroke (70q0 increased 51·7%, with mean age at death 2·7 years lower than expected); and breast cancer (70q0 increased 227·9%, with mean age at death 3·8 years lower than expected). There were six additional causes that showed this same pattern. For females in south Asia, 70q0 due to tracheal, bronchus, and lung cancer increased by 192·6%, with a mean age at death 4·3 years lower than expected (figure 5). In addition, ischaemic heart disease, stroke, breast cancer, chronic kidney disease, and colon and rectum cancer also had increasing 70q0 and lower mean age at death compared with expected age in south Asia. In this super-region, there were also five causes that had an increase in 70q0 only (with mean age at death not lower than expected), and five causes had a mean age at death lower than expected without an increased 70q0. In females in the high-income super-region, only COPD showed both increasing 70q0 and lower mean age at death than expected. Chronic kidney disease, diarrhoeal diseases, and hypertensive heart disease had increasing 70q0, but the mean age at death was not lower than expected. In females in southeast Asia, east Asia, and Oceania, there were only two causes among the leading 20 that had an increased 70q0, breast cancer and diabetes, but both of those causes had mean ages at death below the expected values (figure 5).

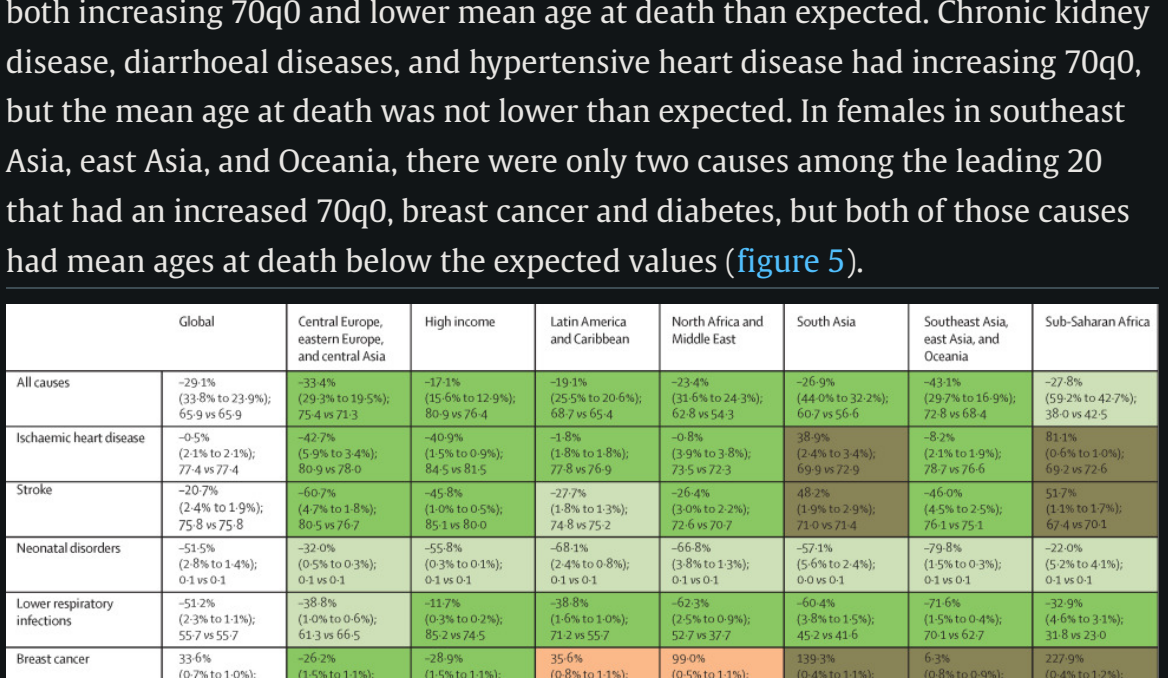
Increasing 70q0, mean age at death lower than expected
Increasing 70q0, mean age at death higher than expected
Decreasing 70q0, mean age at death lower than expected
Decreasing 70q0, mean age at death higher than expected

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Figure 5. Change in 70q0 between 2000 and 2023 and the observed versus expected mean age at death in 2023 for females
The contents of each cell are as follows: percentage change in 70q0 from 2000 to 2023 (70q0 in 2000 to 70q0 in 2023); observed *vs* expected mean age at death in years. 70q0=probability of death before age 70 years.

In males in sub-Saharan Africa, 11 of the top 20 causes of death showed an increase in 70q0 and a mean age that was lower than expected: ischaemic heart disease; stroke; tracheal, bronchus, and lung cancer; cirrhosis and other chronic liver diseases; diabetes; COPD; chronic kidney disease; stomach cancer; colon and rectum cancer; falls; and hypertensive heart disease (figure 6). Additionally, road injuries, self-harm, interpersonal violence, and congenital birth defects had an increase in 70q0, but the mean age at death was not lower than expected. In males in south Asia, eight of the top 20 causes of death showed an increase in 70q0 and a mean age at death that was lower than expected: ischaemic heart disease, stroke, tracheal, bronchus, and lung cancer, COPD, self-harm, chronic kidney disease, stomach cancer, and colon and rectum cancer. Additionally, road injuries, diabetes, falls, and hypertensive heart disease had an increase in 70q0 only. In males in the high-income super-region, 70q0 due to interpersonal violence decreased slightly but had a mean age at death that was more than 3 years lower than expected. Additionally, 70q0 for chronic kidney disease, diarrhoeal diseases, and hypertensive heart disease also increased.

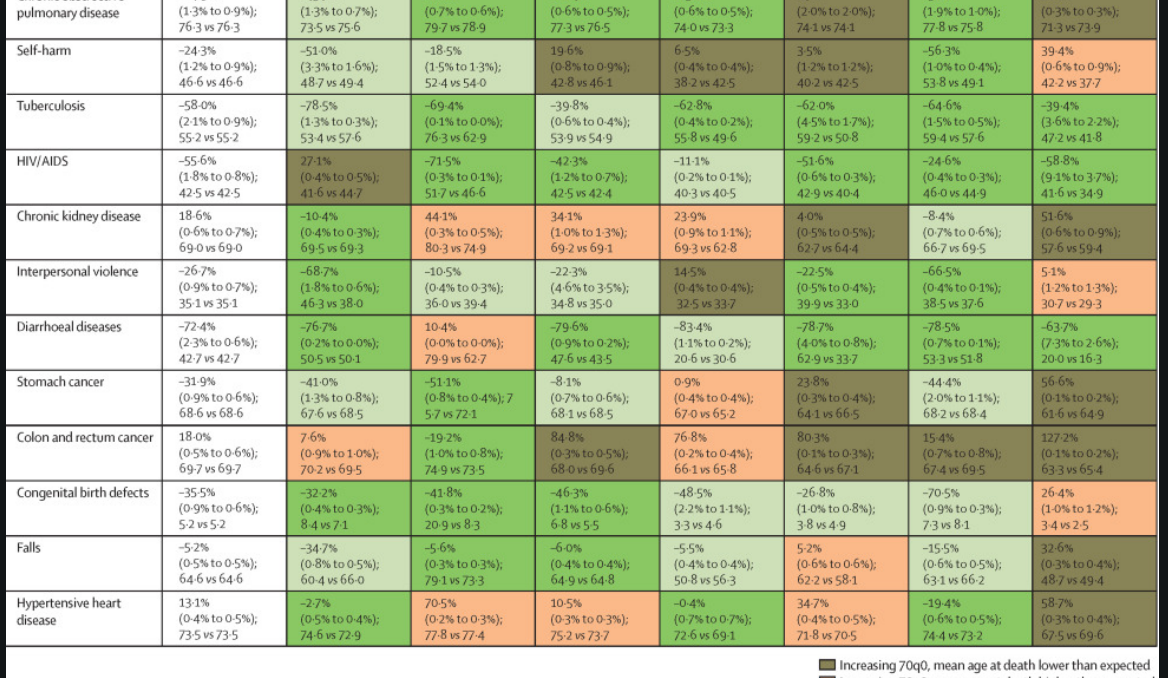
Increasing 70q0, mean age at death lower than expected
Increasing 70q0, mean age at death higher than expected
Decreasing 70q0, mean age at death lower than expected
Decreasing 70q0, mean age at death higher than expected

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Figure 6. Change in 70q0 between 2000 and 2023 and the observed versus expected mean age at death in 2023 for males
The contents of each cell are as follows: percentage change in 70q0 from 2000 to 2023 (70q0 in 2000 to 70q0 in 2023); observed *vs* expected mean age at death in years. 70q0=probability of death before age 70 years.

Discussion

High-level overview on causes of death

This study offers valuable new insights into global causes of human mortality over the past several decades, building upon and expanding previous iterations of GBD research. In a post-COVID-19-pandemic world, our study highlights several encouraging patterns in global health. Across many leading causes of death, there were declines in overall mortality within the period studied, despite disrupted rankings during the height of the COVID-19 pandemic. The rate of total YLLs have also reduced considerably relative to 1990, and particularly for many of the vaccine-preventable diseases, illustrating successes in reductions to preventable causes of death. Patterns in 70q0 show improvements across the globe, with overall declines observed in every GBD super-region, region, and in most countries. Additionally, the all-cause global mean age at death has been rising, indicating that people are generally dying later in life. While these broad-level improvements show considerable promise—particularly in the aftermath of a pandemic—they sometimes conceal disparities occurring at local levels. Differences by sex, age, and location underscore the complexity of global health progress and the persistent challenges in addressing causes of death. We highlight some of these inequalities below.

The role of international cooperation in reducing deaths and YLLs

Over the past three decades, we found large reductions in age-standardised rates of YLLs for four causes—respiratory infections and tuberculosis, nutritional deficiencies, other infectious diseases, and enteric infections—which had individual declines ranging from 58·9% to 79·0%. This achievement was facilitated by many years of sustained international cooperation with local governments. If we expand this scope to include maternal and neonatal disorders, neglected tropical diseases, malaria, HIV/AIDS, and sexually transmitted infections, and categorise them into two groups—vaccine-preventable diseases and other diseases under international intervention plans—we see that the first group had a 66·5% reduction in YLLs, while the second had a 51·6% decline. These findings highlight the profound impact of sustained funding, vaccination, international collaboration, and targeted support programmes with local governments in addressing public health challenges. Previous studies, including GBD 2021, have highlighted the importance of controlling infectious diseases as a key factor in improving life expectancy.[14] Many of these diseases are concentrated in specific populations and locations, making their control an achievable goal.[14] During a time of uncertainty regarding the future of global health funding, it is essential to maintain these efforts, as discontinuing them could jeopardise the gains made in public health.

COVID-19: challenges to recording and lessons for preparedness

The COVID-19 pandemic produced a challenge to global health that has not been seen in recent history, including the difficulties associated with accurately recording and analysing deaths from a novel pandemic. GBD 2021 estimated COVID-19 mortality using confirmed COVID-19 death estimates and excess mortality, an approach which measured the total toll of the pandemic but had limitations for understanding how many deaths were directly attributable to COVID-19.[14] GBD 2023 addresses the issues related to COVID-19 reporting by applying a method for identifying and correcting misclassified COVID-19 deaths in vital registration data in the years 2020–23. This has allowed us to more accurately quantify the COVID-19 pandemic, as well as to correct for inaccurate spikes in mortality from other causes that were instead misclassified COVID-19 deaths. The applied correction occurs after garbage code redistribution to ensure that any deaths from COVID-19 are correctly identified and that other changes in garbage code practices do not result in cause-specific excess deaths.

The additional 2 years of analysis since GBD 2021 allows for a more complete picture of COVID-19 mortality. We estimated a total of 18·0 million people died from COVID-19 globally between 2019 and 2023. This profound loss of life underscores shortcomings in global health systems and a need to fill critical gaps in preparedness for epidemic diseases. Necessary steps to better prepare for the next pandemic should include strengthening health-care infrastructure, enhancing global surveillance, improving vaccine development and delivery, ensuring equitable access to essential and preventive health services, and improving global collaboration to include data sharing and advanced disease monitoring.[32], [33]

In 2020 and 2021, the global mortality rate from COVID-19 for people younger than 70 years was several times higher than the rates for lower respiratory infections before the pandemic, in 2019. In 2023, the rate of COVID-19 deaths decreased to be lower than the sum of all other lower respiratory infections for those younger than 70 years. A similar pattern holds true for those aged 70 years and older. This suggests that COVID-19's impact on mortality could be comparable to other individual lower respiratory infections in future years, and that health-care systems should prepare for and expect future COVID-19 seasons to be endemic.[34]

Measuring the impact of a pandemic is inherently challenging; we recognise that future estimates might be subject to continued improvements as more data become available. The last pandemic of this nature to occur was the 1918 H1N1 influenza pandemic that killed between 21 million and 100 million people within 2 years.[35], [36] The true death toll of the 1918 influenza pandemic still remains uncertain, and even with improved records, there will likely always be some ambiguity surrounding the COVID-19 estimates as well.

Changes in age-specific mortality rate from 2000 to 2023

With all other factors held constant, the age-specific death rate for teenagers, young adults, and older adults (aged 10–69 years) should decline over time, reflecting improvements in health care, socioeconomic conditions, and public health measures.[7] However, this trend has not been universal across all 21 GBD regions. Only four regions—high-income Asia Pacific, central Asia, east Asia, and south AsiA&Mdash;have had a declining age-specific death rate for both males and females across all 5-year age groups for the three Level 1 GBD causes: CMNN diseases, NCDs, and injuries.

In regions displaying an increase in NCDs, primary drivers vary by region; however, we commonly observed high rates of diabetes and kidney disease, cardiovascular disease, substance use disorders, and neoplasms. There were two regions where the increase in NCDs was primarily driven by an increase in drug use disorders: high-income North America and central Europe. The regions facing large increases in neoplasms were north Africa and the Middle East, southeast Asia, tropical Latin America, the Caribbean, and all four regions of sub-Saharan Africa. Diabetes and kidney disease largely contributed to the increase in central Europe, the Caribbean, central Latin America, and southern sub-Saharan Africa. Addressing these trends requires targeted public health interventions, improved health-care access, and socioeconomic policies to mitigate the underlying risk factors.

Among regions displaying an increase in injuries, there was similar variation by region, although common drivers included increases in self-harm, interpersonal violence, conflict and terrorism, environmental heat and cold exposure, and falls. In north Africa and the Middle East and eastern Europe, the rise in injury-related deaths was primarily due to collective and interpersonal violence, in addition to the earthquake in Türkiye. In central and eastern Europe, heatwaves have been occurring more frequently over the past decade. At the same time, increases in high-income North America, central Latin America, and tropical Latin America were all driven by increasing rates of self-harm.

Deaths from violent causes

Our study shows several noteworthy patterns in deaths from violent causes occurring throughout the world. Trends of interpersonal violence showed global-level improvements with regional heterogeneity. Although global mortality from interpersonal violence has generally declined, the regions most heavily affected have seen worsening trends. The primary drivers associated with deaths from interpersonal violence are highly variable across locations.[37] In some parts of the world, the drug trade fuels deaths from this form of violence by driving territorial conflicts, organised crime, and competition over illicit markets.[37] Deaths from interpersonal violence can also be linked to several important social determinants of health, including adverse childhood experiences, alcohol and drug use disorders, and lack of social support, among others.[38], [39]

Deaths from conflict and terrorism are stochastic in nature and have fluctuated over the past three decades, displaying periods of declines and increases influenced by complex regional dynamics. In recent years, the area of conflict has begun to shift from north Africa and the Middle East to central Europe, eastern Europe, and central Asia, due to the war between Russia and Ukraine. Although the regional mortality rates of north Africa and the Middle East are no longer the highest, we found that Palestine had the highest mortality rate and 70q0 due to conflict and terrorism of any country in the world. These findings align with the recently reported number of fatalities in the Gaza Strip,[40] and an estimated 30-year loss in life expectancy within the first 12 months of the war—a conservative estimate that nearly halves the pre-war life expectancy in Palestine.[41]

Global self-harm rates have been trending downwards since the early 1990s, but this progress conceals spikes in self-harm occurring in some locations.[42] We observed an increase in self-harm in central Latin America, and more moderate increases occurring in Andean Latin America, high-income North America, high-income Asia Pacific, and tropical Latin America. There were, however, declines in self-harm in east Asia, particularly in China, where improved economic and social conditions, along with tailored and specified campaigns to reduce self-harm, have been useful in supporting population wellbeing.[43], [44] We also observed that the mean age at death from self-harm has been increasing globally over the past three decades, a finding that potentially reflects both successes and failures with regard to self-harm prevention.[42] While an increase in mean age at death due to self-harm could signal that intervention strategies tailored to younger groups have yielded improvements, increased deaths in older ages might indicate missed opportunities in addressing risk factors that are more relevant in older age groups, such as social isolation, economic insecurity, and increased chronic illness.[45] Taken together, the findings that self-harm persists as a leading cause of death in young people in several regions while the global mean age at death due to self-harm increases suggest pivotal opportunities for further prevention strategies that must be carefully tailored to the intended demographic.

Interpretation of mean age at death

The mean age at death measure provides a clear, easily interpretable metric for summarising the population affected by a given disease or injury. Interventions at both the individual and population levels vary depending on the age of those affected. For example, strategies to improve health outcomes for ischaemic heart disease in South Sudan, where the mean age at death due to this cause is 61·2 years for females, would be likely to focus on prevention strategies and early detection, whereas strategies to improve health outcomes for the same disease in Switzerland, where the mean age at death is 88·4 for females, might focus more on palliative care and limit treatment options. Identifying who is being affected by a disease using a single, interpretable measure could help policy makers to make informed decisions in complex situations.

There are some challenges when drawing comparisons in the observed mean age at death between populations with different age structures. An older population is likely to have an older mean age at death for a given cause than a population with a younger age structure. For this reason, comparing mean ages at death between locations with different population structures is not a good measure for how well a disease is being treated. To account for this, an evaluation of the expected mean age at death reflecting the given demographic's population structure is needed for comparison between locations.

The difference between the expected and observed mean age at death can reflect important factors and risks, beyond just the age distribution, that vary between and within different locations. When the observed mean age at death is lower than expected, it shows that people are dying younger than global rates would suggest. These differences across countries underscore notable inequalities. Causes of death that strongly correlate with SDI and differences between expected and observed mean ages at death are indicative of areas in which the global community has the capacity to improve health outcomes—yet resources and interventions remain unevenly distributed across locations. A lower mean age at death compared with the expected indicates weakness in public health, particularly in preventive measures, early diagnosis, and timely treatment that could delay or prevent deaths. Many examples of lower mean age at death are seen in cardiovascular diseases, cancers, and chronic respiratory diseases in low-income regions, suggesting challenges in both prevention and treatment.

Probability of death

We included the probability of death between the ages of 0 and 70 years (70q0) in this study to assess the likelihood that an individual born today will die from a specific cause before reaching 70 years of age, assuming that current age-specific mortality rates remain unchanged. The 70q0 measure incorporates competing risks, acknowledging that individuals might die from other causes before reaching the high-risk ages for a particular cause. In other words, our goal was to provide a more comprehensive assessment of the 70q0 from a specific cause over the entire lifespan, assuming survival from other causes.

In 2024, the Global Health 2050 report set a target to cut global premature mortality in half by the year 2050, a goal referred to as 50 by 50.[5] In support of this target, we aimed to better position GBD to provide the current state of national premature mortality estimates across causes and locations, which could be useful to track progress on future developments. We used an analysis of 70q0 to detail substantial sources of health loss that most contribute to premature death before age 70 years, providing a roadmap to help countries address their primary contributors to premature mortality.

Several studies suggest that probability of death is a valuable indicator, and providing 70q0 by cause of death highlights areas in which countries can improve the observed age at death for specific causes, drawing attention to locations and causes that have not kept pace with global progress in cause-specific mortality.[5], [46] The probability of death measure also illustrates global and regional success stories in which, as a global health community, we have successfully reduced mortality rates for specific causes in people younger than 70 years. For instance, lower respiratory infections declined 51·2% among females and 47·4% among males between 2000 and 2023 due to reductions in the case-fatality rate and various risk factors,[47] including reductions in household air pollution, a decrease in the prevalence of childhood wasting, and improved vaccine coverage, all of which were effective in reducing the burden of lower respiratory infections.[48] Vaccines against *Haemophilus influenzae* type b and *Streptococcus pneumoniae* are particularly crucial in the reduction of lower respiratory infections in 70q0.[49] Global improvements were also observed in diarrhoeal diseases, with declines of over 70% in 70q0 for males and females. There have been many multisectoral approaches that have contributed to a reduction in diarrhoeal deaths globally in 70q0, many of which have focused on diarrhoeal deaths in children younger than 5 years. These interventions include oral rehydration therapy, enhanced water, sanitation, and hygiene infrastructure, and the rollout of the rotavirus vaccination.[50]

The global reduction of 70q0 due to tuberculosis from 2000 to 2023 is another success story. There was an overall decline of 60·0% in females and 58·0% in males, and notable improvements in some super-regions, particularly central Europe, eastern Europe, and central Asia, where a 78·5% decrease among males and a 71·8% decrease among females occurred in that period. The analysis of 70q0 is more optimistic than other literature on tuberculosis, because the slowest progress in tuberculosis mortality has been in older adults.[47] While we have seen success in reducing 70q0 from tuberculosis, with improvements in mortality for those aged 15 years and younger, more work is needed to reduce tuberculosis mortality in individuals aged 50 years and older,[47], [51] and to reach WHO's End TB Strategy.[52] Further reductions to tuberculosis-related risk factors, such as smoking, alongside early diagnosis and treatment, and the development of less toxic and shorter-duration tuberculosis treatments, are crucial for continued improvements to reach WHO targets by 2035.[47] Lastly, improvements in 70q0 for neonatal disorders globally, decreasing by 51·5% for males and females, reflect the success of efforts to reduce mortality in those younger than 5 years across health sectors and multilaterally. This is generally cited as one of public health's biggest achievements of the 20th century.[53] There are many lessons to be learned from this progress, including the importance of public health standards and measures adopted globally, such as the rollout of the *S pneumoniae* vaccine to prevent lower respiratory infections and the rotavirus vaccine for diarrhoea prevention. Sustainable Development Goal Target 3.2, which aims to end preventable deaths of newborns and children younger than 5 years by 2030, will build on the progress already made in reducing neonatal mortality and support ongoing efforts toward an 80% reduction in tuberculosis cases by 2030, as measured by WHO.[54]

Unfortunately, not all causes of death show an optimistic picture in terms of 70q0. There remain large disparities by cause and location. First, as the epidemiological transition continues, we see rising probabilities of deaths from NCDs in sub-Saharan Africa, Latin America and the Caribbean, south Asia, and southeast Asia, east Asia, and Oceania. With global progress in 70q0, there are outliers where increased mortality rates have occurred during this period. Ten countries saw an increase in 70q0 across all causes between 2000 and 2023; the largest increase was in Palestine at 33·1%, more than double that of the next-largest increase in Lebanon with 15·3%. The staggering increase in Palestine is driven almost entirely by the conflict between Israel and Palestine, with an increase in 70q0 of 8980% due to conflict and terrorism. Despite a substantial rise in 70q0 due to conflict and terrorism in many countries, including the remainder of the top ten countries (Sudan, Ukraine, Russia, Burkina Faso, Myanmar, Israel, Somalia, Syria, and Yemen), none of these countries had an overall increase in their all-cause 70q0 due to their increased risk of conflict deaths.

Patterns in NCD mortality

Rising NCDs worldwide, especially in low-income areas, will represent a significant global health challenge moving forward.[55], [56] Historically, low-income countries have been disproportionately affected by the burden of infectious diseases, but shifts towards more chronic conditions are a reflection of the ongoing global epidemiological transition.[57] In 1990, the three regions with the highest overall mortality rates from all causes were western, eastern, and central sub-Saharan Africa, where 73·4% of deaths came from CMNN diseases. By 2023, CMNN diseases in these regions dropped to 51·4% of all deaths, representing a 30·0% decrease from 1990. Our findings also show that age-standardised mortality rates and 70q0 for both cardiovascular diseases and neoplasms are increasing in sub-Saharan Africa and in south Asia (appendix 2 figure S5). As further reductions in communicable diseases continue, it is likely that deaths from these NCDs could become the dominant sources of mortality in future years.

Findings from our study are in agreement with many studies drawing attention to the surge in NCDs occurring in low-income settings.[57], [58] Although the concept of the epidemiological transition is not new, the speed and scale of the rise in NCDs in low-income regions is increasingly concerning.[57] There are several important implications for health systems when disease burdens transition from communicable diseases to those from non-communicable sources.[10], [57] Health-care infrastructure might face a range of growing challenges associated with increased care needs for chronic disease management requiring long-term care and ongoing treatment. Low-resourced locations remain poorly equipped to address the rising burden of NCDs, with health-care systems often underfunded and unable to provide adequate preventive care or treatment options.[57] Collaborative and focused efforts—including coordinated policy initiatives and prevention programmes targeting key risk factors—are needed to alleviate immediate health challenges related to the rising burden of NCDs in low-income regions and to achieve long-term improvements in global health outcomes.

Limitations

As with any study of this scope, there are several important limitations to consider. We provide cause-specific limitations for every GBD cause of death in detail in appendix 1 (section 3). Here, we highlight limitations with applicability across many causes. First, accuracy of cause-of-death estimates can be affected by data sparsity or unreliability from some regions, time periods, or age groups. In locations for which we have scarce or unreliable data, estimates are interpolated from neighbouring regional patterns by relying on predictive covariates. Second, the cause-of-death estimates rely on medically verifiable sources of cause-of-death data, for which quality can vary. Some datasets do not cover all deaths in a given age, sex, location, and year, and some have high levels of garbage-coded underlying causes of death, which require redistribution algorithms to correct. For transparency about data quality, we publish a star rating of the quality of all vital registration and verbal autopsy data (a 1–5 score compiled based on percentage completeness and percentage garbage). These scores are available in appendix 1 and in a publicly available visualisation tool ↗. Third, for causes with limited data, it is preferable to provide estimates with appropriate uncertainty, rather than providing no information. Fourth, reporting lags in medically verifiable cause-of-death data are a factor in data availability for recent years, particularly 2023; therefore, estimates for these years rely more heavily on the modelling process. Fifth, there are several limitations that pertain to our COVID-19 estimates. While GBD 2023 reflects the most comprehensive set of COVID-19 estimates published by GBD to date, we still have a limited availability of time series for some locations from 2020 to 2023, particularly for 2023. Some location-cause-age-sex groups have a small enough sample size that their time series are stochastic by default, making the development of a counterfactual model difficult. To our knowledge, estimates from GBD 2023 reflect the best account of COVID-19 and miscoded COVID-19 to date. However, as we learn more about the virus and its presentation, it is possible our corrections will be updated to reflect new knowledge in the field. Sixth, mean age at death calculations also have limitations, as they are not standardised for different population age structures. Aggregate estimates are therefore influenced by the most populated areas. As a result, it can be unclear whether the increase in the mean age at death is attributed to a reduction of deaths in younger ages, or if it is simply a result of an ageing population. Our calculation of mean age at death is also limited by the granularity of GBD results. Here, each death is assigned an age group, whereas in reality, each death occurred at a specific age. This strategy does not capture effects within age groups, and it does not show how cohorts age from year to year. Seventh, 70q0 is a broad age group that does not capture improvements made in younger ages if the death occurs before age 70 years. For example, if the mean age at death improved from 30 years to 50 years in a period of time, but the overall mortality rate remained the same, 70q0 would not show this improvement. Lastly, data for stochastic events such as natural disasters and conflicts are generally reported without age and sex detail and instead leverage age-sex splitting using the available detailed data to split the deaths into the granular GBD age groups. These types of events are particularly subject to a lag in reporting, and these estimates will continue to be improved in the future.

Conclusion

GBD cause of death studies are fundamental for understanding mortality trends and aligning them with public health decision making. While progress has been made in reducing deaths from infectious diseases on a global scale, the rising burden of NCDs presents new challenges, particularly for low-income nations. Patterns in premature mortality across the globe have been changing, signifying priority areas for public health intervention. Findings from GBD 2023 show a crucial need for continued investment in health care, improved data collection, and targeted interventions to address both emerging and persistent health issues. Tackling the global health challenges of the future will require sustained international collaboration in the prevention and treatment of both communicable and non-communicable diseases. Strengthening access and quality of health care in low-income and middle-income countries is needed for improving the prevention and treatment of NCDs in particular, which continue to rise as major health threats. A unified global effort will also be necessary to combat the growing number of deaths from drug use and violence, both of which require comprehensive strategies for prevention, treatment, and support. By fostering greater international cooperation and focusing on these key areas, we can make significant progress towards reducing global mortality rates and improving health outcomes for populations worldwide.

GBD 2023 Causes of Death Collaborators

Mohsen Naghavi,* Hmwe Hmwe Kyu,* Bhoomadevi A, Mohammad Amin Aalipour, Hasan Aalruz, Hazim S Ababneh, Bedru J Abafita, Ukachukwu O Abaraogu, Cristiana Abbafati, Madineh Abbasi, Faezeh Abbaspour, Hedayat Abbastabar, Abdallah H A Abd Al Magied, Samar Abd ElHafeez, Ashraf Nabiel Abdalla, Mohammed Altigani Abdalla, Emad M Abdallah, Barkhad Aden Abdeeq, Nadin M I Abdel Razeq, Ahmed Abdelrahman Abdelgalil, Reda Abdel-Hameed, Michael Abdelmasseh, Mahmoud Abdelnabi, Wael M Abdel-Rahman, Arman Abdous, Mostafa M Abdrabou, Jeza Muhamad Abdul Aziz, Deldar Morad Abdulah, Auwal Abdullahi, Toufik Abdul-Rahman, Habtamu Abebe Getahun, Aidin Abedi, Armita Abedi, Parisa Abedi, Asrat Agalu Abejew, Roberto Ariel Abeldaño Zuñiga, Shehab Uddin Al Abid, Syed Hani Abidi, Alemwork Abie, Olugbenga Olusola Abiodun, Richard Gyan Aboagye, Shady Abohashem, Hassan Abolhassani, Ulric Sena Abonie, Nagah M Abourashed, Mohamed Abouzid, Dmitry Abramov, Lucas Guimarães Abreu, Dariush Abtahi, Rana Kamal Abu Farha, Fuad Hamdi A Abuadas, Aminu Kende Abubakar, Nermeen Abu-Elala, Eman Abu-Gharbieh, Sawsan Abuhammad, Ahmad Y Abuhelwa, Hana J Abukhadijah, Niveen ME Abu-Rmeileh, Salahdein Aburuz, Dina Abushanab, Manfred Mario Kokou Accrombessi, Anirudh Balakrishna Acharya, Apurba Acharya, Ousman Adal, Lisa C Adams, Abdu A Adamu, Isaac Yeboah Addo, Oluwafemi Atanda Adeagbo, Tajudeen Adesanmi Adebisi, Isaac Akinkunmi Adedeji, Kamoru Ademola Adedokun, Oluwatobi E Adegbile, Nurudeen A Adegoke, Olumide Thomas Adeleke, Bulcha Guye Adema, Bashir Aden, Isaac Ayodeji Adesina, Miracle Ayomikun Adesina, Juliana Bunmi Adetunji, Habeeb Omoponle Adewuyi, Temitayo Esther Adeyeoluwa, Mache Tsadik Adhana, Ripon Kumar Adhikary, Usha Adiga, Tanin Adl Parvar, Mohd Adnan, Qorinah Estiningtyas Sakilah Adnani, Prince Owusu Adoma, Leticia Akua Adzigbli, David Adzrago, Giuseppina Affinito, Ahmed M Afifi, Clifford Afoakwah, Aanuoluwapo Adeyimika Afolabi, Rotimi Felix Afolabi, Vlad-Adrian Afrăsânie, Saira Afzal, Gizachew Beykaso Agafari, Suneth Buddhika Agampodi, Thilini Chanchala Agampodi, Navidha Aggarwal, Mahdi Aghaalikhani, Sepehr Aghajanian, Seyed Mohammad Kazem Aghamir, Feleke Doyore Agide, César Agostinis Sobrinho, Anurag Agrawal, Williams Agyemang-Duah, Mahsa Ahadi, Bright Opoku Ahinkorah, Aqeel Ahmad, Danish Ahmad, Faisal Ahmad, Ijaz Ahmad, Khabir Ahmad, Khurshid Ahmad, Sajjad Ahmad, Tauseef Ahmad, Waqas Ahmad, Negar Sadat Ahmadi, Ali Ahmed, Ayman Ahmed, Gasha Salih Ahmed, Haroon Ahmed, Junaid Ahmed, Luai A Ahmed, MD Faisal Ahmed, Mehrunnisha Sharif Ahmed, Meqdad Saleh Ahmed, Muktar Beshir Ahmed, Mushood Ahmed, Shabbir Ahmed, Sindew Mahmud Ahmed, Syed Anees Ahmed, Gulzhanat Aimagambetova, Marjan Ajami, Budi Aji, Hossein Akbarialiabad, Saeid Akbarifard, Oluwasefunmi Akeju, Roland Eghoghosoa Akhigbe, Ruslan Akhmedullin, Olufemi Ambrose Akinkuotu, Mohammed Ahmed Akkaif, Wole Akosile, Ashley E Akrami, Ralph Kwame Akyea, Alaa Al Amiry, Salah Al Awaidy, Syed Mahfuz Al Hasan, Ammar Al Homsi, Mohammad Khaled Al Nawayseh, Omar Al Omari, Zain Al Ta'ani, Yazan Al Thaher, Omar Ali Mohammed Al Zaabi, Mohammad Ahmmad Mahmoud Al Zoubi, Mousa Ali Al-Abbadi, Tariq A Alalwan, Ziyad Al-Aly, Khurshid Alam, Manjurul Alam, Mohammad Khursheed Alam, Mostafa Alam, Rasmieh Mustafa Al-Amer, Abebaw Alamrew, Amani Alansari, Turki M Alanzi, Fahmi Y Al-Ashwal, Mohammed Albashtawy, Wafa A Aldhaleei, Mohammed S Aldossary, Robert W Aldridge, Shereen M Aleidi, Bezawit Abeje Alemayehu, Fentahun Alemnew, Melaku Birhanu Alemu, Kefyalew Addis Alene, Ayman Al-Eyadhy, Ali M Alfalki, Fahad D Algahtani, Abdelazeem M Algammal, Khairat Al-Habbal, Nma Bida Alhaji, Samar Al-Hajj, Fadwa Naji Alhalaiqa, Mohammed Khaled Al-Hanawi, Khalid A Alhasan, Ashraf Alhumaidi, Fahad A Alhumaydhi, Amjad Ali, Haroon Muhammad Ali, Irfan Ali, Liaqat Ali, Maratab Ali, Mohammad Daud Ali, Mohammed Usman Ali, Rafat Ali, Shahid Ali, Syed Shujait Ali, Waad Ali, Gianfranco Alicandro, Montaha Al-Iede, Sheikh Mohammad Alif, Hamid Alinejad Rokny, Morteza Alipour, Samah W Al-Jabi, Mohamad Aljofan, Moath Saleh Aljohani, Syed Mohamed Aljunid, Ahmad Alkhatib, Mustafa Alkhawam, Peter Allebeck, Khaled S Allemailem, Mohammed Z Allouh, Wesam Taher Almagharbeh, Sabah Al-Marwani, Nihad A Almasri, Joseph Uy Almazan, Hesham M Al-Mekhlafi, Omar Almidani, Amr Almobayed, Khaldoon Aied Alnawafleh, Hasan Yaser Alniss, Margret Beaula Alocious Sukumar, Mahmoud A Alomari, Mohammad R Alosta, Jaber S Alqahtani, Saleh A Alqahtani, Mohammad R Alqudimat, Ahmad Rajeh Al-Qudimat, Intima Alrimawi, Sahel Majed Alrousan, Salman Khalifah Al-Sabah, Mohammed A Alsabri, Zaid Altaany, Awais Altaf, Alaa B Al-Tammemi, Jaffar A Al-Tawfiq, Malik A Althobiani, Khalid A Altirkawi, Javier Alvarez-Galvez, Nelson Alvis-Guzman, Nelson J Alvis-Zakzuk, Hassan Alwafi, Mohammad Al-Wardat, Yaser Mohammed Al-Worafi, Hany Aly, Mohammad Sharif Ibrahim Alyahya, Hosam Alzahrani, Karem H Alzoubi, Adel Sharaf Al-Zubairi, Ekiyor Joseph Amafah, Joy Amafah, Reza Amani-Beni, Faten Amer, Bardia Amidi, Amr Amin, Tarek Tawfik Amin, Alireza Amindarolzarbi, Saeed Amini, Ehsan Amini-Salehi, Majid Aminzare, Sohrab Amiri, Joanne O Amlag, Dickson A Amugsi, Ganiyu Adeniyi Amusa, Filippos Anagnostakis, Roshan A Ananda, Nazanin Anaraki, Robert Ancuceanu, Deanna Anderlini, David B Anderson, Nguyen Hoang Anh, Abdul-Azeez Adeyemi Anjorin, Samuel Egyakwa Ankomah, Kabilan Annadurai, Sumbul Ansari, Alireza Ansari-Moghaddam, Catherine M Antony, Ernoiz Antriyandarti, Boluwatife Stephen Anuoluwa, Iyadunni Adesola Anuoluwa, Saeid Anvari, Saleha Anwar, Sumadi Lukman Anwar, Razique Anwer, Shahnawaz Anwer, Anayochukwu Edward Anyasodor, Francis Appiah, Juan Pablo Arab, Hossein Arabi, Jalal Arabloo, Mosab Arafat, Daniel T Araki, Aleksandr Y Aravkin, Demelash Areda, Getnet Mesfin Aregu, Jorge Arias de la Torre, Ghazal Arjmand, Benedetta Armocida, Johan Ärnlöv, Jesu Arockiaraj, Mahwish Arooj, Anton A Artamonov, Ashokan Arumugam, Deepavalli Arumuganainar, Umesh Raj Aryal, Nurila Aryntayeva, Mahsa Asadi Anar, Muhammad Asaduzzaman, Syed Mohammed Basheeruddin Asdaq, Mulusew Andualem A Asemahagn, Mulu Tiruneh Asemu, Saeed Asgary, Mohammad Asghari-Jafarabadi, Syed Amir Ashraf, Tahira Ashraf, Mitra Ashrafi, Milad Ashrafizadeh, Bernard Kwadwo Yeboah Asiamah-Asare, Saeed Aslani, Yuni Asri, Batyrbek Assembekov, Seyyed Shamsadin Athari, Alok Atreya, Julie Alaere Atta, Zeenah A Atwan, Khursheed Aurangzeb, Marcel Ausloos, Abolfazl Avan, Núbia Carelli Pereira Avelar, Sana Javaid Awan, Adedapo Wasiu Awotidebe, Lemessa Assefa A Ayana, Haleh Ayatollahi, Yusuf Oloruntoyin Ayipo, Seyed Mohammad Ayyoubzadeh, Sina Azadnajafabad, Arian Azadnia, James Mba Azam, Alireza Azarboo, Zelalem Nigussie Azene, Gulrez Shah Azhar, Amirali Azimi, Farya Azimi, Mohd Yusmaidie Aziz, Sadat Abdulla Aziz, Amin Azizan, Ahmed Y Azzam, Giridhara Rathnaiah Babu, Youngoh Bae, Arvind Bagga, Nasser Bagheri, Sara Bagheri, Elahe Baghizadeh, Fereshteh Baghizadeh, Sana Baghizadeh, Khlood K Baghlaf, Najmeh Bahmanziari, Ruhai Bai, Mohamed Ibrahem Baklola, Abdulaziz T Bako, Wondu Feyisa Balcha, Maher Balkis, Jose Balmori-de-la-Miyar, Mohammadreza Balooch Hasankhani, Ovidiu Constantin Baltatu, Soham Bandyopadhyay, Palash Chandra Banik, Noel C Barengo, Suzanne Lyn Barker-Collo, Hiba Jawdat Barqawi, Amadou Barrow, Sandra Barteit, Lingkan Barua, MD Abu Bashar, Shahid Bashir, Guido Basile, Rehana Basri, Quique Bassat, Mohammad-Mahdi Bastan, Abdul-Monim Batiha, Kavita Batra, Matteo Bauckneht, Mahdis Bayat, Mohammad Amin Bayat Tork, Thomas Beaney, Neeraj Bedi, Narasimha M Beeraka, Massimiliano Beghi, Jina Behjati, Bezawit K Bekele, Almaz Nibret Belay, Demeke Mesfin Belay, Asnake Gashaw Belayneh, Melesse Belayneh, Abel Cherkos Belete, Gokce Belge Bilgin, Muhammad Bashir Bello, Olorunjuwon Omolaja Bello, Umar Muhammad Bello, Luis Belo, Apostolos Beloukas, Riyad Bendardaf, Isabela M Bensenor, Samiun Nazrin Bente Kamal Tune, Maria Bergami, Alemshet Yirga Berhie, Abiye Assefa Berihun, Amiel Nazer C Bermudez, Robert S Bernstein, Gregory J Bertolacci, Paola Bertuccio, Paulo J G Bettencourt, Ajeet Singh Bhadoria, Akshaya Srikanth Bhagavathula, Neeraj Bhala, Buna Bhandari, Kayleigh Bhangdia, Charmi Bhanushali, Nikha Bhardwaj, Pankaj Bhardwaj, Ashish Bhargava, Sonu Bhaskar, Anup Bhat, Priyadarshini Bhattacharjee, Shuvarthi Bhattacharjee, Gurjit Kaur Bhatti, Jasvinder Singh Bhatti, Mohiuddin Ahmed Bhuiyan, Zulfiqar A Bhutta, Soumitra S Bhuyan, Haoran Bi, Sibhatu Kassa Biadgilign, Raluca Bievel-Radulescu, Naif Kandash Binsaleh, Catherine Bisignano, Atanu Biswas, Bijit Biswas, Mohammad Shahangir Biswas, Ahmad Naoras Bitar, Molalegne Bitew, Bruno Bizzozero-Peroni, Tone Bjørge, Virginia Bodolica, Eyob Ketema Bogale, Lucimere Bohn, Obasanjo Afolabi Bolarinwa, Paria Bolourinejad, Aime Bonny, Sri Harsha Boppana, Hamed Borhany, Mina Borran, Sudipta Bose, Samuel Adolf Bosoka, Alejandro Botero Carvajal, Soufiane Boufous, Christopher Boxe, Dejana Braithwaite, Luisa C Brant, Michael Brauer, Nicholas J K Breitborde, Susanne Breitner, Hermann Brenner, Edmond D Brewer, Maria L Bringas Vega, Julie Brown, Annie J Browne, Traolach Brugha, Raffaele Bugiardini, Norma B Bulamu, Tsion Samuel Bunare, Danilo Buonsenso, Richard A Burns, Akeem Olayinka Busari, Felix Busch, Yasser Bustanji, Nadeem Shafique Butt, Zahid A Butt, Sanjay C J, Tianji Cai, Rose Cairns, Mehtap Çakmak Barsbay, Daniela Calina, Luis Alberto Cámera, Luciana Aparecida Campos, Ismael Campos-Nonato, Fan Cao, Si Cao, Angelo Capodici, Rosario Cárdenas, Giulia Carreras, Juan Jesus Carrero, Andrea Carugno, Andre F Carvalho, Felix Carvalho, Márcia Carvalho, Ana Paula Carvalho-e-Silva, Joao Mauricio Castaldelli-Maia, Carlos A Castañeda-Orjuela, Giulio Castelpietra, Ferrán Catalá-López, Alberico L Catapano, Maria Sofia Cattaruzza, Luca Cegolon, Francieli Cembranel, Muthia Cenderadewi, Kelly M Cercy, Ester Cerin, Pamela Roxana Chacón-Uscamaita, Chiranjib Chakraborty, Sandip Chakraborty, Joht Singh Chandan, Rama Mohan Chandika, Miyuru Chandradasa, Baskaran Chandrasekaran, Vijay Kumar Chattu, Victoria Chatzimavridou-Grigoriadou, Anis Ahmad Chaudhary, Sirshendu Chaudhuri, Akhilanand Chaurasia, An-Tian Chen, Catherine S Chen, Guangjin Chen, Haiyan Chen, Hana Chen, Haowei Chen, Hui Chen, Rucheng Chen, Shanquan Chen, Simiao Chen, Xiang Chen, Haojin Cheng, Ka Ching Cheung, Nicholas WS Chew, Gerald Chi, Fatemeh Chichagi, Izumi Chihara, Odgerel Chimed-Ochir, Patrick R Ching, Jesus Lorenzo Chirinos-Caceres, Daniel Youngwhan Cho, William C S Cho, Bryan Chong, Yuen Yu Chong, Hou In Chou, Enayet Karim Chowdhury, Sreshtha Chowdhury, Hanne Christensen, Ting-Wu Chuang, Isaac Sunday Chukwu, Erin Chung, Sheng-Chia Chung, Sunghyun Chung, Muhammad Chutiyami, Arrigo Francesco Giuseppe Cicero, Cain C T Clark, Fred Cohen, Alyssa Columbus, Joao Conde, Stephen E Congly, Nathalie Conrad, Leslie Trumbull Cooper, Alexandru Corlateanu, Samuele Cortese, Paolo Angelo Cortesi, Claudia Cosma, Ewerton Cousin, Emma Johnson Cowart, Michael H Criqui, Andrew Crist, 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Olum, Bolajoko Olubukunola Olusanya, Jacob Olusegun Olusanya, Oluwafemi G Oluwole, Folorunsho Bright Omage, Goran Latif Omer, Abidemi E Omonisi, Kanyin Liane Ong, Sandersan Onie, Obinna E Onwujekwe, Oluwaseyi Aina Gbolade Opesemowo, John Nelson Opio, Marcel Opitz, Aksoltan Shyhdurdyevna Oradova, Michal Ordak, Verner N Orish, Raffaele Ornello, Atakan Orscelik, Alberto Ortiz, Esteban Ortiz-Prado, Augustus Osborne, Samuel M Ostroff, John W Ostrominski, Uchechukwu Levi Osuagwu, Olayinka Osuolale, Elham H Othman, Adrian Otoiu, Abdu Oumer, Jerry John Ouner, Amel Ouyahia, Mayowa O Owolabi, Irene Amoakoh Owusu, Oladayo Ayobami Oyebanji, Kolapo Oyebola, Tope Oyelade, Kehinde Adewole Oyeniran, Oyetunde T Oyeyemi, Ilker Ozsahin, Mahesh P A, Kevin Pacheco-Barrios, Alicia Padron-Monedero, Jagadish Rao Padubidri, Dimpal Manilal Paija, Anton Pak, Yeganeh Pakbaz, Pramod Kumar Pal, Tamás Palicz, Raffaele Palladino, Tejasri Paluvai, Feng Pan, Sujogya Kumar Panda, Songhomitra Panda-Jonas, Deepshikha Pande Katare, Seithikurippu R Pandi-Perumal, Victoria Pando-Robles, Apurvakumar Pandya, Helena Ullyartha Pangaribuan, Georgios D Panos, Leonidas D Panos, Ioannis Pantazopoulos, Anca Pantea Stoian, Giovanni Paolino, Mario Virgilio Papa, Ilias Papadimopoulos, Paraskevi Papadopoulou, Peyvand Parhizkar Roudsari, Romil R Parikh, Chulwoo Park, Seoyeon Park, Arpit Parmar, Roberto Passera, Jay Patel, Mitesh Patel, Neel Navinkumar Patel, Sangram Kishor Patel, Satyananda Patel, Bharat Smita Umakant Patil, Shankargouda Patil, Dimitrios Patoulias, Apurba Patra, Venkata Suresh Patthipati, Shrikant Pawar, Shubhadarshini Pawar, Hamidreza Pazoki Toroudi, Neil Pearce, Amy E Peden, Paolo Pedersini, Jarmila Pekarcikova, Louise Penberthy, Veincent Christian Filipino Pepito, Emmanuel K Peprah, Prince Peprah, João Perdigão, Gavin Pereira, Gladymar Perez Chacon, Arokiasamy Perianayagam, Norberto Perico, Simone Perna, Konrad Pesudovs, Pavlo Petakh, Ionela-Roxana Petcu, Olumuyiwa James Peter, Fanny Emily Petermann-Rocha, William A Petri, Hoang Nhat Pham, Hoang Tran Pham, Tung Thanh Pham, Anil K Philip, Michael R Phillips, Zahra Zahid Piracha, Edoardo Pirera, Moein Piroozkhkah, Saeed Pirouzpanah, Enrico Pisoni, Evgenii Plotnikov, Indrashis Podder, Dimitri Poddighe, Roman V Polibin, Ramesh Poluru, Arjun Pon Avudaiappan, Ville T Ponkilainen, Ion Popa, Djordje S Popovic, Thantrira Porntaveetus, Sajjad Pourasghary, Reza Pourbabaki, Farzad Pourghazi, Naeimeh Pourtaheri, Sergio I Prada, Jalandhar Pradhan, Rifky Octavia Pradipta, Akila Prashant, Elton Junio Sady Prates, Natalie Pritchett, Harsh Priya, Nicola Riccardo Pugliese, Jagadeesh Puvvula, Nameer Hashim Qasim, Ibrahim Qattea, Xiang Qi, Zhipeng Qi, Yanan Qiao, Zahiruddin Syed Quazi, Navid Rabiee, Reza Rabiei, Basuki Rachmat, Raghu Anekal Radhakrishnan, Venkatraman Radhakrishnan, Maja R Radojčić, Negar Radpour, Hadi Raeisi Shahraki, Lida Rafati, Ibrar Rafique, Pracheth Raghuveer, Fakher Rahim, Hawbash Mohammed-Amin Rahim, Sajjad Rahimi, Vafa Rahimi-Movaghar, Fryad Majeed Rahman, Mahbubur Rahman, Md Mosfequr Rahman, Mohammad Hifz Ur Rahman, Mohammad Meshbahur Rahman, Mosiur Rahman, Amir Masoud Rahmani, Saeed Rahmani, Masoud Rahmati, Ghasem Rahmatpour Rokni, Hakim Rahmoune, Diego Raimondo, Ivano Raimondo, Sunil Kumar Raina, Jeffrey Pradeep Raj, Adarsh Raja, Sathish Rajaa, Erta Rajabi, Gunaseelan Rajendran, Judah Rajendran, Vinoth Rajendran, Shaman Rajindrajith, Pushp Lata Rajpoot, Prashant Rajput, Mahmoud Mohammed Ramadan, Majed Ramadan, Kadar Ramadhan, Chitra Ramasamy, Shakthi Kumaran Ramasamy, Zahra Ramezani, Marzieh Ramezani Farani, Robinson Ramírez-Vélez, Juwel Rana, Kirtan Rana, Shailendra Singh Rana, Chhabi Lal Ranabhat, Nemanja Rancic, Smitha Rani, Fatemeh - Ranjbar Noei, Chythra R Rao, Kumuda Rao, Mithun Rao, Davide Rasella, Sina Rashedi, Vahid Rashedi, Mamunur Rashid, Mohammad-Mahdi Rashidi, Mohammad Aziz Rasouli, Ashkan Rasouli-Saravani, Azad Rasul, Devarajan Rathish, Abdur Rauf, Santosh Kumar Rauniyar, Ilari Rautalin, Ramin Ravangard, David Laith Rawaf, Lal Rawal, Reza Rawassizadeh, Bahman Razi, C Mahony Reategui-Rivera, Elrashdy Redwan, Aqeeb Ur Rehman, Faizan Ur Rehman, Wajiha Rehman, Lennart Reifels, Rainer Reile, Giuseppe Remuzzi, Bhageerathy Reshmi, Stefano Restaino, Luis Felipe Reyes, Mina Rezaei, Nazila Rezaei, Nima Rezaei, Mohsen Rezaeian, Donya Rezazadeh Eidgahi, Taeho Gregory Rhee, Yohanes Andy Rias, Antonio Luiz P Ribeiro, Tércia Moreira Ribeiro da Silva, Jennifer Rickard, Moattar Raza Rizvi, Hannah Elizabeth Robinson-Oden, Hermano Alexandre Lima Rocha, João Rocha Rocha-Gomes, Mónica Rodrigues, Thales Philipe Rodrigues da Silva, Jefferson Antonio Buendia Rodriguez, Leonardo Roever, Peter Rohloff, Iftitakhur Rohmah, Susanne Röhr, David Rojas-Rueda, Megan L Rolfzen, Debby Syahru Romadlon, Michele Romoli, Luca Ronfani, Kevin T Root, Emily Rosenblad, Amirhossein Roshanshad, Morteza Rostamian, Gregory A Roth, Kunle Rotimi, Himanshu Sekhar Rout, Hanieh Rouzbahani, Reza Rouzbahani, Jemma V Rowlands, Adrija Roy, Bedanta Roy, Priyanka Roy, Sharmistha Roy, Shubhanjali Roy, Simanta Roy, Parameswari Royapuram Parthasarathy, Enrico Rubagotti, Susan Fred Rumisha, Michele Russo, Godfrey Mutashambara Rwegerera, Aly M A Saad, Michela Sabbatucci, Maha Mohamed Saber-Ayad, Siamak Sabour, Perminder S Sachdev, Seyed Kiarash Sadat Rafiei, Basema Ahmad Saddik, Bashdar Abuzed Sadee, Tarannom Sadegh, Ehsan Sadeghi, Erfan Sadeghi, Fatemeh Sadeghi-Ghyassi, Mohd Saeed, Umar Saeed, Maryam Saeedi, Mahdi Safdarian, Sare Safi, Sher Zaman Safi, Rajesh Sagar, Mastooreh Sagharichi, Amene Saghazadeh, Dominic Sagoe, Indranil Saha, Nondo Saha, Fatemeh Saheb Sharif-Askari, Narjes Saheb Sharif-Askari, Amirhossein Sahebkar, Kirti Sundar Sahu, Zahra Saif, S Mohammad Sajadi, Md Refat Uz Zaman Sajib, Mirza Rizwan Sajid, Dorsa Salabat, Payman Salamati, Luciane B Salaroli, Mohamed A Saleh, Leili Salehi, Mahdi Salehi, Marwa Rashad Salem, Mohammed Z Y Salem, Aanuoluwa James Salemcity, Dauda Salihu, Sohrab Salimi, Malik Sallam, Hossein Samadi Kafil, Jayami Eshana Samaranayake, Saad Samargandy, Waqas Sami, Yoseph Leonardo Samodra, Abdallah M Samy, Sandeep G Sangle, Elaheh Sanjari, Sathish Sankar, Francesco Sanmarchi, Francesca Sanna, Damian F Santomauro, Itamar S Santos, Lucas H C C Santos, Milena M Santric-Milicevic, Adekunle Sanyaolu, Bruno Piassi Sao Jose, Krishna Prasad Sapkota, Sivan Yegnanarayana Iyer Saraswathy, Yaser Sarikhani, Hemen Sarma, Mohammad Sarmadi, Gargi Sachin Sarode, Sachin C Sarode, Benn Sartorius, Arash Sarveazad, Michele Sassano, Mukesh Kumar Sathya Narayanan, Maheswar Satpathy, Reza Sattarpour, Davide Sattin, Mehrdad Savabi Far, Monika Sawhney, Sangeeta Gopal Saxena, Ganesh Kumar Saya, Abu Sayeed, Christophe Schinckus, Jurgen Carlo Schmidt, Maria Inês Schmidt, Rachel D Schneider, Art Schuermans, Austin E Schumacher, Aletta Elisabeth Schutte, Ghil Schwarz, David C Schwebel, Falk Schwendicke, Sneha Annie Sebastian, Amin Sedigh, Soraya Seedat, Mario Šekerija, Muthamizh Selvamani, Vimalraj Selvaraj, Yuliya Semenova, Mohammad H Semreen, Fikadu Waltengus Sendeku, Pallav Sengupta, Yigit Can Senol, Subramanian Senthilkumaran, Sadaf G Sepanlou, Edson Serván-Mori, Yashendra Sethi, Seyed Mohammad Seyed Alshohadaei, Allen Seylani, Abubakar Sha'aban, Mahan Shafie, Arezoo Shafieioun, Shazlin Shaharudin, Muhammad Shahbaz, Samiah Shahid, Syed Ahsan Shahid, Endrit Shahini, Fatemeh Shahrahmani, Hamid R Shahsavari, Moyad Jamal Shahwan, Masood Ali Shaikh, Alireza Shakeri, Ali Shakerimoghaddam, Ali S Shalash, Muhammad Aaqib Shamim, Farzane Shams, Mehran Shams-Beyranvand, Anas Shamsi, Alfiya Shamsutdinova, Dan Shan, Shan Shan, Mohd Shanawaz, Amin Sharifan, Javad Sharifi Rad, Avimanu Sharma, Bhoopesh Kumar Sharma, Bunty Sharma, Gaurav Sharma, Kamal Sharma, Kamlesh Sharma, Manoj Sharma, Ravi Kumar Sharma, Ujjawal Sharma, Vishal Sharma, Shamee Shastry, Maryam Shayan, Babangida Shehu Bappah, Fateme Sheida, Ali Sheidaei, Ali Sheikhy, Rekha Raghuveer Shenoy, Samendra P Sherchan, B Suresh Kumar Shetty, Shiran Shetty, Fanchao Shi, Fang Shi, Amir Shiani, Belayneh Fentahun Shibesh, Kenji Shibuya, Desalegn Shiferaw, Tariku Shimels, Md Monir Hossain Shimul, Min-Jeong Shin, Rahman Shiri, Reza Shirkoohi, Aminu Shittu, Abdul-karim Olayinka Shitu, Ivy Shiue, Velizar Shivarov, Nathan A Shlobin, Ambreen Shoaib, Shayan Shojaei, Sina Shool, Seyed Afshin Shorofi, Sunil Shrestha, Suleiman Adeiza Shuaibu, Kerem Shuval, Zahra Siavashpour, Nicole Remaliah Samantha Sibuyi, Emmanuel Edwar Siddig, Ahmed Kamal Siddiqi, Diego Augusto Santos Silva, João Pedro Silva, Luís Manuel Lopes Rodrigues Silva, Padam Prasad Simkhada, Biagio Simonetti, Abhinav Singh, Amit Singh, Balbir Bagicha Singh, Baljinder Singh, Bhim Pratap Singh, Harmanjit Singh

Singh, Balbir Bagicha Singh, Baljinder Singh, Bhim Pratap Singh, Harmanjit Singh, Harpreet Singh, Jasvinder A Singh, Jawahar Singh, Kalpana Singh, Mayank Singh, Narinder Pal Singh, Paramdeep Singh, Poornima Suryanath Singh, Puneetpal Singh, Rakesh K Singh, Samer Singh, Satwinder Singh, Surendra Singh, Surjit Singh, Mukesh Kumar Sinha, Robert Sinto, Sarah Brooke Sirota, Dagne Feleke Siyoum, Natia Skhvitaridze, Anna Aleksandrovna Skryabina, David A Sleet, Mahdieh SobhZahedi, Marzieh Soheili, MdSalman Sohel, Somaye Sohrabi, Shipra Solanki, Lencho Kajela Solbana, Solikhah Solikhah, Sameh S M Soliman, Weiyi Song, Aayushi Sood, Prashant Sood, Soroush Soraneh, Reed J D Sorensen, Joan B Soriano, Fernando Sousa, Marco Aurelio Sousa, Ireneous N Soyiri, Ceren Soylu, Michael Spartalis, Chandrashekhar T Sreeramareddy, Suresh Kumar Srinivasamurthy, Shyamkumar Sriram, Prateek Srivastav, Devin Bailey Srivastava, Lauryn K Stafford, Jeffrey D Stanaway, Muhammad Haroon Stanikzai, Nadine Steckling-Muschack, Dan J Stein, Caitlyn Steiner, Jaimie D Steinmetz, Paschalis Steiropoulos, Blossom Christa Maree Stephan, Aleksandar Stevanović, Leo Stockfelt, Sebastian Straube, Jacob L Stubbs, Peter Stubbs, Omer Subasi, Narayan Subedi, Alisha Suhag, Hasnat Sujon, Thitiporn Sukaew, Surajo Kamilu Sulaiman, Auwal Garba Suleiman, Muritala Suleiman Odidi, Muhammad Suleman, Mark J M Sullman, Anusha Sultan Meo, Haitong Zhe Sun, Jing Sun, Mao-ling Sun, Xiaodong Sun, Xiaohui Sun, Zhong Sun, Zhuanlan Sun, Suraj Sundaragiri, Thanigaivel Sundaram, Johan Sundström, David Sunkersing, Sumam Sunny, Vinay Suresh, Hani Susianti, Chandan Kumar Swain, Vivianne M Swart, Dayinta Annisa Syaiful, Tasmin L Symons, Lukasz Szarpak, Mindy D Szeto, Sree Sudha T Y, Payam Tabaee Damavandi, Rafael Tabarés-Seisdedos, Fatemeh Sadat Tabatabaei, Seyed Shahaboddin Tabatabaei, Seyyed Mohammad Tabatabaei, Seyed-Amir Tabatabaeizadeh, Shima Tabatabai, Celine Tabche, Mohammad Tabish, Takahiro Tabuchi, Getu Ferenji Tadesse, Farzad Taghizadeh-Hesary, Zanan Mohammed-Ameen Taha, Jabeen Taiba, Shima Tajabadi, Iman M Talaat, Mircea Tampa, Jacques Lukenze Tamuzi, Ker-Kan Tan, Mohammad Tanashat, Haosu Tang, Mohsan Tanveer, Abiyu Abadi Tareke, Sarvenaz Taridashti, Ingan Ukur Tarigan, Mengistie Kassahun Tariku, Saba Tariq, Aigul Yelgondiyevna Tazhiyeva, Tarilate Temedie-Asogwa, Mohamad-Hani Temsah, Masayuki Teramoto, Azimeraw Arega Tesfu, Nahom Worku Teshager, Gizachew A Tessema, Jay Tewari, Alireza Teymouri, Chandan Kumar Thakur, Kavumpurathu Raman Thankappan, Rekha Thapar, Ismaeel Tharwat, Samar Tharwat, Rasiah Thayakaran, Muthu Thiruvengadam, Manuel Sebastian Thomas, Wei Tian, Jansje Henny Vera Ticoalu, Madi Tleshev, Sojit Tomo, Marcello Tonelli, Roman Topor-Madry, Mathilde Touvier, Marcos Roberto Tovani-Palone, Khaled Trabelsi, Quynh Thuy Huong Tran, Tam Quoc Minh Tran, Thang Huu Tran, Nguyen Tran Minh Duc, Domenico Trico, Indang Trihandini, Manjari Tripathi, Tulika Tripathi, Samuel Joseph Tromans, Quynh Xuan Nguyen Truong, Thien Tan Tri Tai Truyen, Gary Tse, Vasilis-Spyridon Tseriotis, Evangelia Eirini Tsermpini, Lorainne Tudor Car, Munkhtuya Tumurkhuu, Zhouting Tuo, Biruk Shalmeno Tusa, Sok Cin Tye, Stefanos Tyrovolas, Aniefiok John Udoakang, Atta Ullah, Himayat Ullah, Saeed Ullah, Muhammad Umair, Hauwa Onozasi Umar, Lawan Umar, Muhammad Umar, Muhammad Umar, Shehu Salihu Umar, Eduardo A Undurraga, Bhaskaran Unnikrishnan, Dinesh Upadhya, Era Upadhyay, Dipan Uppal, Daniele Urso, Jibrin Sammani Usman, Kelechi Julian Uzor, Hande Uzunçıbuk, Pratyusha Vadagam, Asokan Govindaraj Vaithinathan, Pascual R Valdez, Mario Valenti, Zahir Vally, Jef Van den Eynde, Javad Varasteh, Joe Varghese, Pavani Varma, Tommi Juhani Vasankari, Sampara Vasishta, Srivatsa Surya Vasudevan, Alireza Vaysi, Siavash Vaziri, Narayanaswamy Venketasubramanian, Madhur Verma, Megan Verma, Poonam Verma, Massimiliano Veroux, Georgios-Ioannis Verras, Simone Vidale, Mathavaswami Vijayageetha, Simone Villa, Jorge Hugo Villafañe, Leonardo Villani, David Villarreal-Zegarra, Francesco S Violante, Senthil Visaga Ambi, Luciano Magalhães Vitorino, Vasily Vlassov, Stein Emil Vollset, Avina Vongpradith, Theo Vos, Mehdi Vosoughi, Elpida Vounzoulaki, Linh Vu, Isidora S Vujcic, Krishna Dhavan Vyas, Henok Toga Wada, Yasir Waheed, Mohd Wahid, Mugi Wahidin, Mandaras Tariku Walde, Megha Walia, Jin-Yi Wan, Arvinder Wander, Fang Wang, Fulin Wang, Junshi Wang, Liang Wang, Qingzhi Wang, Ruixuan Wang, Shu Wang, Wanzhou Wang, Xing Wang, Xuequan Wang, Yan Wang, Yanzhong Wang, Yichen Wang, Youxin Wang, Yuan-Pang Wang, Zhihua Wang, Tanveer A Wani, Mary Njeri Wanjau, Ahmed Bilal Waqar, Muhammad Waqas, John W Ward, Paul Ward, Toyiba Hiyaru Wassie, Stefanie Watson, Ishanka Weerasekara, Fei-Long Wei, Xueying Wei, Robert G Weintraub, Daniel J Weiss, Eli J Weiss, Katherine M Wells, Andrea Werdecker, Ronny Westerman, Taweewat Wiangkham, Yohanes Cakrapradipta Wibowo, Dakshitha Praneeth Wickramasinghe, Nuwan Darshana Wickramasinghe, Samuel Wiebe, Angga Wilandika, Peter Willeit, Shadrach Wilson, Andrew Awuah Wireko, Charles Shey Wiysonge, Abay Tadesse Woday, Bogdan Wojtyniak, Nathnael Abera Woldehana, Dawit Habte Woldeyes, Axel Walter Wolf, Tewodros Eshete Wonde, Yen Jun Wong, Daniel Tarekegn Worede, Abdulhalik Workicho, Minichil Chanie Worku, Ai-Min Wu, Chenkai Wu, Felicia Wu, James Fan Wu, Jinyi Wu, Peng Wu, Zenghong Wu, Yihun Miskir Wubie, Ratna Dwi Wulandari, Zhijia Xia, Guangqin Xiao, Lishun Xiao, Na Xiao, Wanqing Xie, Site Xu, Suowen Xu, Xiaoyue Xu, Yvonne Yiru Xu, Mukesh Kumar Yadav, Vikas Yadav, Mahnaz Yadollahi, Saba Yahoo (Syed), Galal Yahya, Kazumasa Yamagishi, Guangcan Yan, Haibo Yang, Yuichiro Yano, Haiqiang Yao, Laiang Yao, Amir Yarahmadi, Habib Yaribeygi, Haya Yasin, Mohamed A Yassin, Yuichi Yasufuku, Sanni Yaya, Pengpeng Ye, Meghdad Yeganeh, Ali Cem Yekdeş, Mohammad Hossein YektaKooshali, Kuanysh A Yergaliyev, Subah Abderehim Yesuf, Saber Yezli, Siyan Yi, Dehui Yin, Paul Yip, Malede Berihun Yismaw, Yazachew Engida Yismaw, Dong Keon Yon, Naohiro Yonemoto, Seok-Jun Yoon, Mustafa Z Younis, Saideh Yousefi, Abdilahi Yousuf, Chuanhua Yu, Yong Yu, Hui Yuan, Faith H Yuh, Ghazala Yunus, Umar Yunusa, Siddhesh Zadey, Vesna Zadnik, Mubashir Zafar, Manijeh Zaghampour, Emilia Zainal Abidin, Fathiah Zakham, Nazar Zaki, Giulia Zamagni, Nelson Zamora, Hussaini Zandam, Aurora Zanghì, Heather J Zar, Kourosh Zarea, Mohammed Zawiah, Mohammed G M Zeariya, Abay Mulu Zenebe, Sebastian Zensen, Nejimu Biza Zepro, Eyael M Zeru, Tiansong Zhan, Yongle Zhan, Beijian Zhang, Casper J P Zhang, Haijun Zhang, Julio Min Fei Zhang, Kexin Zhang, Liqun Zhang, Meixin Zhang, Xiaoyi Zhang, Xiu-Hang Zhang, Yunquan Zhang, Zhiqiang Zhang, Sholpan Bolatovna Zhangelova, Hanqing Zhao, Jianhui Zhao, Jiefeng Zhao, Yang Zhao, Zhongyi Zhao, Anthony Zhong, Claire Chenwen Zhong, Jiayan Zhou, Juexiao Zhou, Bin Zhu, Abzal Zhumagaliuly, Magdalena Zielińska, Ghazal Zoghi, Mohamed Ali Zoromba, Zhiyong Zou, Rafat Mohammad Zrieq, Liesl J Zuhlke, Lilik Zuhriyah, Alimuddin Zumla, Ahed H Zyoud, Sa'ed H Zyoud, Shaher H Zyoud, Eve E Wool†, and Christopher J L Murray†. *Joint first authors †Joint senior authors

Affiliations

Please see appendix 4 (pp 13–67) for the affiliations for individual authors.

Contributors

Please see appendix 4 (pp 67–90) for more detailed information about individual author contributions to the research, divided into the following categories: managing the overall research enterprise; writing the first draft of the manuscript; primary responsibility for applying analytical methods to produce estimates; primary responsibility for seeking, cataloguing, extracting, or cleaning data; designing or coding figures and tables; providing data or critical feedback on data sources; developing methods or computational machinery; providing critical feedback on methods or results; drafting the manuscript or revising it critically for important intellectual content; and managing the estimation or publications process. The corresponding author and first author had access to and verified the data. The corresponding author confirms all authors have seen and approved the final text.

Data sharing

For detailed information on data sources and estimates, please visit the GHDx GBD 2023 website at http://ghdx.healthdata.org/gbd-2023 ↗.

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The International Coordinator of the 1997/98, 2001/02, 2005/06 and 2009/10 surveys was Prof Candace Currie and the Data Bank Manager for the 1997/98 survey was Prof Bente Wold, whereas for the following survey Prof Oddrun Samdal was the Databank Manager. A list of principal investigators in each country can be found at http://www.hbsc.org. This paper uses data from the WHO Study on global AGEing and adult health (SAGE). This study is based on data from Eurostat, Latvia European Health Interview Survey 2008 and 2014. The responsibility for all conclusions drawn from the data lies entirely with the author(s). This study is based on data from Eurostat, Romania European Health Interview Survey 2008 and 2014. The responsibility for all conclusions drawn from the data lies entirely with the author(s). This paper uses data from the Moldova 2013 and 2021 STEPS surveys, implemented by Ministry of Health (Moldova) with the support of WHO. 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Data for this research were accessed via the Irish Social Science Data Archive (www.ucd.ie/issda). The original creators bear no responsibility for analysis or interpretation of them. This analysis is based on the Statistics Canada Canadian Community Health Survey Microdata File which contains anonymised data collected in the 2013–2014 Canadian Community Health Survey. All computations, use and interpretation of these data are entirely that of IHME. This paper uses data from the Kenya 2015 STEPS survey, implemented by Kenya National Bureau of Statistics, Ministry of Health (Kenya) with the support of WHO. This analysis uses data or information from the LASI Pilot micro data and documentation. The development and release of the LASI Pilot Study was funded by the National Institute on Ageing / National Institute of Health (R21AG032572, R03AG043052, and R01 AG030153). 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The CRELES project (Costa Rican Longevity and Healthy Aging Study) is a longitudinal study by the University of Costa Rica's Centro Centroamericano de Población and Instituto de Investigaciones en Salud, in collaboration with the University of California at Berkeley. The original pre-1945 cohort was funded by the Wellcome Trust (grant 072406), and the 1945–1955 Retirement Cohort was funded by the U.S. National Institute on Aging (grant R01AG031716). The study Principal Investigators are Luis Rosero-Bixby and William H. Dow, and co-Principal Investigators Xinia Fernández and Gilbert Brenes. This paper uses data from the Myanmar 2014 STEPS survey, implemented by Ministry of Health (Myanmar) with the support of WHO. HBSC is an international study carried out in collaboration with WHO/EURO. The International Coordinator of the 2013/2014 surveys was Prof Candace Currie and the Data Bank Manager was Prof Oddrun Samdal. A list of principal investigators in each country can be found at http://www.hbsc.org. The Canada Health Measures Survey 2016–2017 contains information licensed under the Open Government License Canada. This research used information from the Health Surveys for epidemiological surveillance of the Undersecretary of Public Health. The author thanks the Ministry of Health of Chile, having allowed them to have access to the database. All the results obtained from the study or research are the responsibility of the author and in no way compromise that institution. In this paper use is made of data of the DNB Household Survey administered by Centerdata (Tilburg University, The Netherlands). Those who carried out the original collection and analysis of the Jamaica Survey of Living Conditions bear no responsibility for their further analysis or interpretation. 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This study is based in part on data from Eurostat, European Union Labor Force Survey, 1992–2016. The responsibility for all conclusions drawn from the data lies entirely with the authors. This study is based on data from Eurostat, Belgium Health Interview Survey 2008 and 2009. The responsibility for all conclusions drawn from the data lies entirely with the author(s). This study is based on data from Eurostat, Cyprus Health Interview Survey 2008–2009. The responsibility for all conclusions drawn from the data lies entirely with the author(s). This study is based on data from Eurostat, Czech Republic European Health Interview Survey 2006–2009 and 2013–2015. The responsibility for all conclusions drawn from the data lies entirely with the author(s). This study is based on data from Eurostat, Estonia European Health Interview Survey 2006–2007 and 2014. The responsibility for all conclusions drawn from the data lies entirely with the author(s). 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VACS data are owned by the Government of Cote d'Ivoire and made available by the Centers for Disease Control and Prevention through a Data Use Agreement. This paper uses data from the Mongolia 2019 STEPS survey, implemented by the Ministry of Health, Public Health Institute with the support of WHO. This paper uses data from the Jordan 2019 STEPS survey, implemented by the Ministry of Health with the support of WHO. This paper uses data from the Turkmenistan 2018 STEPS survey, implemented by the Ministry of Health and Medical Industry with the support of WHO. This study is based on data from Eurostat, Austria European Health Interview Survey 2006–2007 and 2013–2015. The responsibility for all conclusions drawn from the data lies entirely with the author(s). This study is based on data from Eurostat, Belgium European Health Interview Survey 2013 and 2013–2015. The responsibility for all conclusions drawn from the data lies entirely with the author(s). 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The accuracy of the authors' statistical analysis and the findings they report are not the responsibility of ECDC. ECDC is not responsible for the conclusions or opinions drawn from the data provided. ECDC is not responsible for the correctness of the data and for data management, data merging and data collation after provision of the data. ECDC shall not be held liable for improper or incorrect use of the data.

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